

Diazepam: Drug information - UpToDate

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For additional information see "Diazepam: Patient drug information" and "Diazepam: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

FDA Extends Expiration Dates for Certain Meridian Autoinjectors: Updated July 2017

The FDA is alerting health care providers and emergency responders that certain lots of *AtroPen* (atropine), *CANA* (diazepam), morphine sulfate, and pralidoxime chloride autoinjectors manufactured by Meridian Medical Technologies can be used beyond the labeled expiration date, which should help mitigate potential shortages of these drugs. To ensure patient safety, products should be stored under the manufacturer's labeled storage conditions.

Additional information may be found at <http://www.fda.gov/Drugs/DrugSafety/ucm376367.htm>.

ALERT: US Boxed Warning

Risks from concomitant use with opioids:

Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of these drugs for patients for whom alternative treatment options are inadequate. Limit dosages and durations to the minimum required. Follow patients for signs and symptoms of respiratory depression and sedation.

Abuse, misuse, and addiction:

Buccal film: The buccal film (Libervant) is approved for use in pediatric patients 2 to 5 years of age. The unapproved use of the buccal film exposes users to risks of abuse, misuse, and addiction, which can lead to overdose or death. Abuse and misuse of benzodiazepines commonly involve concomitant use of other medications, alcohol, and/or illicit substances, which is associated with an increased frequency of serious adverse outcomes.

Injection, oral, nasal spray, rectal gel: The use of benzodiazepines, including diazepam, exposes users to risks of abuse, misuse, and addiction, which can lead to overdose or death. Abuse and misuse of benzodiazepines commonly involve concomitant use of other medications, alcohol, and/or illicit substances, which is associated with an increased frequency of serious adverse outcomes. Before prescribing diazepam and throughout treatment, assess each patient's risk for abuse, misuse, and addiction.

Dependence and withdrawal reactions:

Buccal film: The continued use of benzodiazepines may lead to clinically significant physical dependence. The risks of dependence and withdrawal increase with longer treatment duration and higher daily dose. Although the buccal film is indicated only for intermittent use, if used more frequently than recommended, abrupt discontinuation or rapid dosage reduction of the buccal film may precipitate acute withdrawal reactions, which can be life-threatening. For patients using the buccal film more frequently than recommended, to reduce the risk of withdrawal reactions, use a gradual taper to discontinue diazepam.

Oral: The continued use of benzodiazepines, including diazepam, may lead to clinically significant physical dependence. The risks of dependence and withdrawal increase with longer treatment duration and higher daily dose. Abrupt discontinuation or rapid dosage reduction of diazepam after continued use may precipitate acute withdrawal reactions, which can be life-threatening. To reduce the risk of withdrawal reactions, use a gradual taper to discontinue diazepam or reduce the dosage.

Injection, nasal spray, rectal gel: The continued use of benzodiazepines, including diazepam, may lead to clinically significant physical dependence. The risks of dependence and withdrawal increase with longer treatment duration and higher daily dose. Although diazepam is indicated only for intermittent use, if used more frequently than recommended, abrupt discontinuation or rapid dosage reduction may precipitate acute withdrawal reactions, which can be life-threatening. For patients using diazepam more frequently than recommended, to reduce the risk of withdrawal reactions, use a gradual taper to discontinue diazepam.

Brand Names: US

- Diastat AcuDial [DSC];
- Diastat Pediatric [DSC];
- diazePAM Intensol;
- Libervant [DSC];
- Valium;
- Valtoco 10 MG Dose;
- Valtoco 15 MG Dose;
- Valtoco 20 MG Dose;
- Valtoco 5 MG Dose

Brand Names: Canada

- Diastat;
- Valium

Pharmacologic Category

- Antiseizure Agent, Benzodiazepine;
- Benzodiazepine

Dosing: Adult

Dosage guidance:

Safety: Reduce dose or avoid use in patients receiving opioids or with significant chronic disease (eg, respiratory compromise). Avoid use in patients with a history of substance use, misuse of medications, or depression, except for acute or emergency situations (eg, status epilepticus) (Ref).

Dosage form information: Libervant buccal film has been discontinued in the United States for >1 year.

Alcohol withdrawal syndrome

Alcohol withdrawal syndrome:

Note: Withdrawal will progress at different rates in some patients; flexibility in dosing and duration is warranted (Ref). Regimens vary and depend on withdrawal history, degree of current withdrawal symptoms, blood alcohol concentration, and whether the patient is treated inpatient or in the ambulatory setting. Many facilities only treat alcohol withdrawal in the ambulatory setting if Clinical Institute Withdrawal Assessment for Alcohol, revised scale (CIWA-Ar) score is ≤ 15 (mild to moderate

alcohol withdrawal) and there is no history of withdrawal seizures or delirium tremens (Ref). Some experts recommend avoiding IM administration due to variable absorption (Ref). The following are two suggested regimens.

Symptom-triggered regimen:

Note: Some experts recommend inpatient treatment and symptom-triggered dosing for patients with CIWA-Ar >15 (moderate to severe alcohol withdrawal) (Ref).

IV, Oral: 5 to 10 mg as needed per institution-specific protocol until appropriate sedation achieved; dose and frequency determined by withdrawal symptom severity using a validated severity assessment scale, such as the CIWA-Ar (Ref).

Fixed-dose regimen :

Note: Some experts recommend fixed-dose regimens for patients with CIWA-Ar ≤15 (mild to moderate alcohol withdrawal) (Ref). In addition to scheduled doses shown in the example regimen below, provide up to 5 additional as-needed 10 mg doses to be used over the 4-day treatment period for breakthrough symptoms (Ref).

Day 1: **Oral, IV:** 10 mg every 6 hours (Ref).

Day 2: **Oral, IV:** 10 mg every 8 hours (Ref).

Day 3: **Oral, IV:** 10 mg every 12 hours (Ref).

Day 4: **Oral, IV:** 10 mg at bedtime, then discontinue diazepam (Ref).

Anxiety

Anxiety:

Anxiety, acute/severe (monotherapy or adjunctive therapy):

IM, IV, Oral: 2 to 10 mg every 3 to 6 hours as needed up to 40 mg/day; adjust dose based on response and tolerability (Ref).

Anxiety disorders (monotherapy or adjunctive therapy) (alternative agent):

Note: Generally used short term for symptom relief until preferred therapy (eg, serotonin reuptake inhibitor) is effective (eg, 4 to 6 weeks, followed by tapering). Long-term, low-dose (eg, 2.5 mg/day) therapy may be considered in select patients only when other treatments are ineffective or poorly tolerated (Ref). Use with caution in patients with posttraumatic stress disorder; benzodiazepines may cause cognitive changes (Ref).

Initial: **Oral:** 2 to 5 mg once or twice daily; increase gradually based on response and tolerability up to 40 mg/day in 2 to 4 divided doses (Ref).

Procedural anxiety (premedication):

IV: 2 to 10 mg **or** 0.03 to 0.1 mg/kg once (maximum single dose: 10 mg) 5 to 15 minutes prior to procedure; if needed due to incomplete response and/or duration of procedure, may repeat the dose (usually 50% of the initial dose) after 5 to 30 minutes (Ref). **Note:** In obese patients, non-weight-based dosing is preferred (Ref).

Oral (off-label): 2 to 10 mg once 30 to 60 minutes prior to procedure; if needed due to incomplete response, may repeat the dose (usually 50% of the initial dose) after 30 to 60 minutes (Ref).

Hydroxychloroquine/Chloroquine toxicity

Hydroxychloroquine/Chloroquine toxicity (severe) (off-label use):

Note: Use is recommended in patients with severe toxicity (eg, hypotension, QTc prolongation, hypokalemia) in combination with other supportive measures (eg, mechanical ventilation, epinephrine, cardiovascular monitoring) (Ref).

IV: 2 mg/kg once administered over 30 minutes, followed by 1 to 2 mg/kg/day for 2 to 4 days (Ref).

Intoxication

Intoxication (cocaine, methamphetamine, and other sympathomimetics) (off-label use):
Based on limited data.

IV: 2 to 10 mg every 3 to 10 minutes as needed for agitation, sedation, seizures, hypertension, and tachycardia until desired symptom control achieved; doses up to 20 mg may be considered in severe agitation based on response and tolerability. Large, cumulative doses may be required for some patients; monitor for respiratory depression and hypotension. **Note :** If IV access is not possible, consider IM administration; however, IM diazepam time to peak drug levels is slower than IM midazolam (Ref).

Muscle spasm, spasticity, and/or rigidity

Muscle spasm, spasticity, and/or rigidity (alternative agent):

Oral: Initial: 2 mg once or twice daily **or** 5 mg at bedtime; may increase total daily dose in increments of 2 to 5 mg in weekly intervals based on response and tolerability, up to 40 to 60 mg/day in 3 to 4 divided doses (Ref).

Neuroleptic malignant syndrome

Neuroleptic malignant syndrome (adjunctive therapy) (off-label use):

Note: Following withdrawal of causative agent while continuing supportive care, may use for muscle rigidity and/or agitation (Ref).

IV: 10 mg every 8 hours until symptom resolution (Ref).

Opioid withdrawal in the emergency setting

Opioid withdrawal (autonomic instability and agitation) in the emergency setting (alternative agent) (adjunctive therapy) (off-label use):

Note: Based on limited data. Reserve for use when preferred opioid agonists are not available or not preferred by patient (Ref).

IV: 5 to 10 mg every 5 to 10 minutes until hemodynamic stability and adequate sedation achieved (Ref).

Organophosphate poisoning

Organophosphate poisoning (adjunctive therapy) (off-label use):

Note: For treatment of seizures or agitation. Also consider for patients with coma, apnea, or neuromuscular paralysis since benzodiazepines may decrease neurocognitive dysfunction and CNS-mediated respiratory depression (Ref).

IV: 10 mg; may repeat 1 to 2 more injections every 5 to 10 minutes for persistent seizure activity (Ref).

Seizures

Seizures:

Note: If IV access is not available, IM diazepam is not recommended due to erratic absorption and slow time to peak drug levels (IM midazolam is recommended) (Ref).

Acute active seizures (non-status epilepticus):

Intranasal: 0.2 mg/kg as a single dose; may repeat once based on response and tolerability after ≥ 4 hours. Maximum dose: Two doses per episode. Do not use for more than 1 episode every 5 days or more than 5 episodes per month.

The following table (derived from manufacturer labeling) provides acceptable weight ranges for each dose, such that patients will receive between 90% and 180% of the calculated recommended dose.

Recommended Intranasal Diazepam Dosage for Adults

Weight

Dose (rounded from 0.2 mg/kg)

Number of nasal spray devices

Number of sprays

28 to 50 kg

10 mg

One 10 mg device

One spray in one nostril

51 to 75 kg

15 mg

Two 7.5 mg devices

One spray in each nostril

76 kg and up

20 mg

Two 10 mg devices

One spray in each nostril

IV: 5 to 10 mg as a single dose given at a maximum infusion rate of 5 mg/minute; may repeat at 3- to 5-minute intervals if seizures continue up to a total dose of 30 mg (Ref).

Oral (adjunctive therapy) (alternative agent):

Note: Although previously described in literature for use in seizure clusters as rescue therapy, oral diazepam tablets are no longer recommended for this use due to concerns related to involuntary movements, swallowing and risks of aspiration, and delayed gastric emptying. Safer dosage forms are currently recommended (eg, IV, rectal gel, nasal spray, etc.) (Ref).

Manufacturer's labeling: Dosing in prescribing information may not reflect current clinical practice.

Oral: 2 to 10 mg administered 2 to 4 times daily.

Rectal gel (generally for use in prehospital setting): 0.2 mg/kg (round dose up to the nearest 2.5 mg increment; maximum dose: 20 mg) **or** 10 to 20 mg as a single dose (Ref). Maximum duration: 5 episodes/month and 1 episode every 5 days (Ref).

Status epilepticus (alternative agent):

IV: 5 to 10 mg as a single dose given at a maximum infusion rate of 5 mg/minute; may repeat dose at 3- to 5-minute intervals if seizures continue. A nonbenzodiazepine antiseizure agent should follow to prevent seizure recurrence, even if seizures have ceased (Ref).

Rectal gel (generally for use in prehospital setting) (off-label): 0.2 to 0.5 mg/kg (round dose up to the nearest 2.5 mg increment; maximum dose: 20 mg) as a single dose (Ref).

Serotonin syndrome

Serotonin syndrome (serotonin toxicity) (off-label use):

IV: 5 to 10 mg every 8 to 10 minutes until symptoms resolve (Ref).

Vertigo, acute episodes

Vertigo, acute episodes (alternative agent) (off-label use):

Note: Reserve use for symptomatic relief of episodes lasting several hours to days (maximum duration: 3 days); chronic use may impede adaptation and recovery (Ref).

IV, Oral: 1 to 5 mg every 12 hours as needed for up to 48 to 72 hours (Ref).

Discontinuation of therapy: Unless safety concerns require a more rapid withdrawal, gradually taper to detect reemerging symptoms and minimize rebound and withdrawal symptoms in patients receiving therapy ≥ 4 weeks or as appropriate based on patient-specific factors (Ref).

Low or moderate dose, no concerns for benzodiazepine use disorder: Taper total daily dose by 20% to 25% every week based on response and tolerability (taper increments will be limited by available dosage forms) (Ref).

Extended or high-dose therapy, or suspected benzodiazepine use disorder: Taper total daily dose by approximately 25% every 1 to 2 weeks based on response, tolerability, and individual patient factors (taper increments will be limited by available dosage forms) (Ref). Reduce dose more rapidly in the beginning, and slow the dose reduction as the taper progresses because earlier stages of withdrawal are easier to tolerate (Ref). The optimal duration and taper increment will vary; up to 6 months may be necessary for some patients on higher doses, and a taper rate of 50% every week may be tolerated in some patients (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: An increased incidence of CNS-related adverse effects has been reported in patients with hypoalbuminemia (which is frequently observed in patients with end-stage kidney disease or critical illness) (Ref).

Altered kidney function: No dosage adjustment necessary for any degree of kidney impairment; use with caution, especially with prolonged courses (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzable (highly protein bound, large V_d): No supplemental dose or dosage adjustment necessary; use with caution, especially with prolonged courses (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzable (highly protein bound, large V_d): No dosage adjustment necessary; use with caution, especially with prolonged courses (Ref).

CRRT: No dosage adjustment necessary; use with caution, especially with prolonged courses (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary; use with caution, especially with prolonged courses (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution because distribution and half-life may increase, and clearance may decrease significantly. The oral tablets are contraindicated in severe hepatic impairment.

Dosing: Older Adult

Note: Avoid use, may be appropriate for seizure disorders, alcohol withdrawal, severe generalized anxiety disorder, or procedural anesthesia (Ref).

Older and/or debilitated patients:

IM, IV: Initial: 2 to 5 mg; increase gradually based on response and tolerability.

Intranasal: Due to the increased half-life in elderly patients, consider reducing dose.

Oral: Initial: 2 to 2.5 mg 1 to 2 times daily; increase gradually based on response and tolerability.

Rectal gel: Due to the increased half-life in elderly and debilitated patients, consider reducing dose.

Dosing: Pediatric

(For additional information see "Diazepam: Pediatric drug information")

Dosage guidance:

Dosage form information: Libervant buccal film has been discontinued in the United States for >1 year.

Seizures, acute

Seizures, acute:

Buccal film (eg, Libervant):

Children 2 to 5 years: Buccal: Dosing varies with weight. May repeat dose in 4 hours; do not exceed 2 doses in 24 hours. Do not repeat dose if patient has difficulty breathing or excessive sedation. Do not exceed maximum treatment frequency of 1 episode **every 5 days** and 5 episodes per month.

Table 1: **Buccal Diazepam Dose for Acute Seizure in Children 2 to 5 Years**

Weight	Dose
6 to <11 kg	5 mg
11 to <16 kg	7.5 mg
16 to <21 kg	10 mg
21 to <26 kg	12.5 mg
26 to 30 kg	15 mg

Intranasal (eg, Valtoco):

Intranasal: Dosing varies with age and weight; younger children require a larger mg/kg dose. May repeat dose in 4 hours; do not exceed 2 doses in 24 hours. Do not repeat dose if patient has difficulty breathing or excessive sedation. Do not exceed maximum treatment frequency of 1 episode **every 5 days** and 5 episodes per month.

Children 2 to <6 years:

Weight-Band Dosage Recommendations for Intranasal Diazepam: Children 2 to <6 Years^a

Weight

Dose (mg)

Quantity and type of nasal device

Number of sprays

^a Weight-band fixed doses are based on a dose of 0.5 mg/kg/dose.

6 to <12 kg

5 mg

One 5 mg device

1 spray in 1 nostril

12 to <23 kg

10 mg

One 10 mg device

1 spray in 1 nostril

23 to <34 kg

15 mg

Two 7.5 mg devices

2 sprays delivered as 1 spray in **each** nostril

Children 6 to <12 years:

Weight-Band Dosage Recommendations for Intranasal Diazepam: Children 6 to <12 Years^a

Weight

Dose (mg)

Quantity and type of nasal device

Number of sprays

^a Weight-band fixed doses are based on a dose of 0.3 mg/kg/dose.

10 to <19 kg

5 mg

One 5 mg device

1 spray in 1 nostril

19 to <38 kg

10 mg

One 10 mg device

1 spray in 1 nostril

38 to <56 kg

15 mg

Two 7.5 mg devices

2 sprays delivered as 1 spray in **each** nostril

56 to 74 kg

20 mg

Two 10 mg devices

2 sprays delivered as 1 spray in **each** nostril

Children ≥12 years and Adolescents:

Weight-Band Dosage Recommendations for Intranasal Diazepam: Children ≥12 Years and Adolescents^a

Weight

Dose (mg)

Quantity and type of nasal device

Number of sprays

^a Weight-band fixed doses are based on a dose of 0.2 mg/kg/dose.

14 to <28 kg

5 mg

One 5 mg device

1 spray in 1 nostril

28 to <51 kg

10 mg

One 10 mg device

1 spray in 1 nostril

51 to <76 kg

15 mg

Two 7.5 mg devices

2 sprays delivered as 1 spray in **each** nostril

≥76 kg

20 mg

Two 10 mg devices

2 sprays delivered as 1 spray in **each** nostril

Rectal gel formulation (eg, Diastat):

Weight-based dosing: Round dose up to the nearest 2.5 mg increment. Do not use more than 5 times per month or for more than 1 episode every 5 days.

Infants and Children 6 months to <2 years: Rectal: Dose not established.

Children 2 to <6 years: Rectal: 0.5 mg/kg; maximum dose: 20 mg/dose; dose may be repeated in 4 to 12 hours if needed.

Children 6 to <12 years: Rectal: 0.3 mg/kg; maximum dose: 20 mg/dose; dose may be repeated in 4 to 12 hours if needed.

Children ≥12 years and Adolescents: Rectal: 0.2 mg/kg; maximum dose: 20 mg/dose; dose may be repeated in 4 to 12 hours if needed.

Weight-band dosing: **Note:** Dosing will provide between 90% and 180% of the calculated weight-based dose; safety has been established in clinical trials. Do not use more than 5 times per month or for more than 1 episode every 5 days.

Children 2 to <6 years:

Table 2: **Weight-Band Dosage Recommendations for Rectal Gel Diazepam: Children 2 to <6 Years**

Weight	Dose (mg)	Second Dose
6 to <11 kg	5 mg	May repeat dose in 4 to 12 hours if needed.
11 to <16 kg	7.5 mg	May repeat dose in 4 to 12 hours if needed.
16 to <21 kg	10 mg	May repeat dose in 4 to 12 hours if needed.
21 to <26 kg	12.5 mg	May repeat dose in 4 to 12 hours if needed.
26 to <31 kg	15 mg	May repeat dose in 4 to 12 hours if needed.
31 to <36 kg	17.5 mg	May repeat dose in 4 to 12 hours if needed.
36 to 44 kg	20 mg	May repeat dose in 4 to 12 hours if needed.

Children 6 to <12 years:

Table 3: **Weight-Band Dosage Recommendations for Rectal Gel Diazepam: Children 6 to <12 Years**

Weight	Dose (mg)	Second Dose
10 to <17 kg	5 mg	May repeat dose in 4 to 12 hours if needed.
17 to <26 kg	7.5 mg	May repeat dose in 4 to 12 hours if needed.
26 to <34 kg	10 mg	May repeat dose in 4 to 12 hours if needed.
34 to <42 kg	12.5 mg	May repeat dose in 4 to 12 hours if needed.
42 to <51 kg	15 mg	May repeat dose in 4 to 12 hours if needed.
51 to <59 kg	17.5 mg	May repeat dose in 4 to 12 hours if needed.
59 to 74 kg	20 mg	May repeat dose in 4 to 12 hours if needed.

Children ≥ 12 years and Adolescents:

Table 4: **Weight-Band Dosage Recommendations for Rectal Gel Diazepam: Children ≥ 12 Years and Adolescents**

Weight	Dose (mg)	Second Dose
14 to <26 kg	5 mg	May repeat dose in 4 to 12 hours if needed.
26 to <38 kg	7.5 mg	May repeat dose in 4 to 12 hours if needed.
38 to <51 kg	10 mg	May repeat dose in 4 to 12 hours if needed.
51 to <63 kg	12.5 mg	May repeat dose in 4 to 12 hours if needed.
63 to <76 kg	15 mg	May repeat dose in 4 to 12 hours if needed.
76 to <88 kg	17.5 mg	May repeat dose in 4 to 12 hours if needed.
88 to 111 kg	20 mg	May repeat dose in 4 to 12 hours if needed.

Rectal solution: Undiluted 5 mg/mL parenteral formulation: Infants, Children, and Adolescents: Reported dosing range: 0.2 to 0.5 mg/kg/dose; may repeat dose; maximum dose: 20 mg/dose (Ref).

Seizures, acute, due to organophosphate poisoning

Seizures, acute, due to organophosphate poisoning: Limited data available (Ref):

IM: Various dosage forms available; not all doses can be delivered.

Infants and Children: IM: 0.2 to 0.5 mg/kg/dose; repeat every 2 to 5 minutes; maximum total dose: 5 mg (Ref). **Note:** Following toxin exposure (organophosphates, carbamates) in children, an autoinjector fixed dose of 10 mg is recommended (Ref).

Adolescents: IM: 5 to 10 mg/dose (Ref).

IM CANA 10 mg autoinjector:

Children: 1 CANA autoinjector (10 mg).

Adolescents: 2 to 3 CANA autoinjectors (20 to 30 mg).

IV/Intraosseous:

Infants and Children <5 years: IV, Intraosseous: 0.2 to 0.5 mg/kg/dose; may repeat dose in 15 to 30 minutes; maximum total dose: 5 mg (Ref).

Children ≥ 5 years and Adolescents: IV, Intraosseous: 0.15 to 0.2 mg/kg/dose; maximum single dose: 10 mg/dose; may repeat every 10 to 15 minutes; maximum total dose: 30 mg total (Ref).

Status epilepticus

Status epilepticus:

IV (preferred route):

Infants >30 days, Children, and Adolescents: IV: 0.15 to 0.2 mg/kg/dose slow IV; may repeat dose once in 5 minutes; maximum dose: 10 mg/dose (Ref).

Rectal/ Note: For use when IV access unavailable.

Children 2 to <6 years: Rectal: 0.5 mg/kg; maximum dose: 20 mg/dose (Ref).

Children 6 to <12 years: Rectal: 0.3 mg/kg; maximum dose: 20 mg/dose (Ref).

Children ≥12 years and Adolescents: Rectal: 0.2 mg/kg; maximum dose: 20 mg/dose (Ref).

Febrile seizure, prophylaxis

Febrile seizure, prophylaxis: Limited data available:

Note: Although there is evidence that prophylaxis with diazepam may reduce recurrence of febrile seizures, use is not routinely recommended since risks of toxicity generally outweigh the benefits (Ref).

Children: Oral: 0.33 mg/kg/dose every 8 hours; initiate therapy at first sign of fever and continue for 24 hours after fever resolves (Ref).

Spasticity/muscle spasms

Spasticity/muscle spasms:

General dosing: **Note:** Initiate therapy with lowest dose; dose should be individualized and titrated to effect and tolerability:

Fixed dosing: Infants ≥6 months, Children, and Adolescents: Oral: Initial: 1 to 2.5 mg 3 to 4 times daily; increase gradually as needed and tolerated (Ref).

Weight-directed dosing:

Infants ≥6 months and Children <12 years: Oral: 0.12 to 0.8 mg/kg/day in divided doses every 6 to 8 hours; maximum dose: 10 mg/dose (Ref).

Children ≥12 years and Adolescents: Oral: 2 to 10 mg 2 to 4 times daily; begin on low end of the dosing range with a once-daily dose at bedtime and titrate (Ref).

Cerebral palsy-associated spasticity: Limited data available.

Note: Dose should be individualized and titrated to effect and tolerability.

Weight-directed dosing: Children and Adolescents: Oral: Usual Initial: 0.04 mg/kg/dose 3 times daily; titrate as needed; median reported maximum daily dose: 0.8 mg/kg/**day** in divided doses 4 times daily or 15 mg/**day** (Ref).

Fixed dosing: Children ≥5 years and Adolescents: Oral: Initial: 1 to 2 mg/dose 3 times daily; may titrate to response (Ref). Doses up to 5 mg 4 times daily have been reported (Ref).

Low-dose fixed bedtime dosing:

Children <12 years:

<8.5 kg: Oral: 0.5 to 1 mg at bedtime (Ref).

8.5 to 15 kg: Oral: 1 to 2 mg at bedtime (Ref).

Tetanus-associated spasm:

Fixed dosing: **Note:** Respiratory support should be available during therapy.

Infants >30 days and children <5 years: IV, IM: 1 to 2 mg every 3 to 4 hours as needed (Ref).

Children ≥5 years and Adolescents: IV, IM: 5 to 10 mg every 3 to 4 hours as needed (Ref).

Weight-directed dosing: Infants, Children, and Adolescents: IV: Initial: 0.1 to 0.2 mg/kg/dose every 2 to 6 hours; titrate as needed. **Note:** Initial recommended adult dose is 5 mg/dose (Ref).

Muscle spasm/spasticity associated with chronic/terminal illness (eg, palliative care settings): Limited data available:

Infants, Children, and Adolescents:

Oral:

Weight-directed dosing: Oral: Initial: 0.05 mg/kg/dose every 6 hours; titrate up to maximum dose: 0.2 mg/kg/dose every 6 hours; maximum dose: 10 mg/dose; reported range: 0.12 to 0.8 mg/kg/day divided every 6 to 12 hours (Ref).

Fixed dosing: Patient weight >50 kg: Oral: 2.5 mg every 6 hours; titrate to maximum dose: 10 mg/dose (Ref).

IM, IV:

Weight-directed dosing: IM, IV: Initial: 0.05 mg/kg/dose every 6 hours; titrate up to maximum dose: 0.2 mg/kg/dose every 6 hours; maximum dose: 10 mg/dose (Ref); reported range: 0.05 to 0.2 mg/kg/dose every 6 to 12 hours; maximum total dose: 0.6 mg/kg cumulative in 8 hours (Ref).

Sedation, anxiolysis, and amnesia prior to procedure

Sedation, anxiolysis, and amnesia prior to procedure: Limited data available:

Oral:

Infants ≥ 6 months: Oral: 0.2 to 0.3 mg/kg 45 to 60 minutes prior to procedure. Maximum dose: 10 mg/dose (Ref).

Children: Oral: 0.2 to 0.5 mg/kg 45 to 60 minutes prior to procedure; maximum dose: 10 mg/dose (Ref).

Adolescents: Oral: 0.2 to 0.3 mg/kg 45 to 60 minutes prior to procedure. Maximum dose: 10 mg/dose (Ref).

IV:

Infants and Children: IV: Initial: 0.05 to 0.1 mg/kg over 3 to 5 minutes, titrate slowly to effect (maximum total dose: 0.25 mg/kg) (Ref).

Adolescents: IV: 5 mg; may repeat with 2.5 mg if needed (Ref).

Continuous spikes and waves during sleep/Electrical status epilepticus in sleep

Continuous spikes and waves during sleep (CSWS)/Electrical status epilepticus in sleep (ESES): High-dose diazepam:

Children and Adolescents: Oral: Usual dose: 1 mg/kg once on the initial night; maximum initial dose: 40 mg/dose; then 0.5 mg/kg/dose at bedtime; maximum maintenance dose: 20 mg/dose; therapy reportedly continued for 1 to 3 months and tapered over 1 to 3 months (Ref). Reported daily dose range: 0.23 to 2.02 mg/kg/day (mean daily dose: 0.99 mg/kg/day) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; based on adult experience and data, no dosage adjustment may be necessary for kidney impairment; use with caution, especially with prolonged courses; monitor closely (Ref); an increased incidence of CNS-related adverse effects has been reported in patients with hypoalbuminemia (which is frequently observed in patients with end-stage kidney disease or critical illness) (Ref).

Hemodialysis: Not dialyzable (0% to 5%); supplemental dose is not necessary.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution; distribution and half-life may increase, and clearance may decrease significantly. The oral tablets are contraindicated in severe hepatic impairment.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions may vary by route of administration.

>10%: Nervous system: Drowsiness (23%)

1% to 10%:

Cardiovascular: Hypotension ($\geq 1\%$), vasodilation (2%)

Dermatologic: Skin rash (3%)

Gastrointestinal: Abdominal pain ($\geq 1\%$), diarrhea (4%), dysgeusia (2%), hiccups ($\geq 1\%$)

Nervous system: Agitation ($\geq 1\%$), asthenia (1%), ataxia (3%), changes in thinking ($\geq 1\%$), confusion ($\geq 1\%$), dizziness (3%), dysarthria ($\geq 1\%$), emotional lability ($\geq 1\%$), euphoria (3%), headache (5%), nervousness ($\geq 1\%$), pain ($\geq 1\%$), speech disturbance ($\geq 1\%$), vertigo ($\geq 1\%$)

Respiratory: Asthma (2%), epistaxis (2%), nasal discomfort (5%), rhinitis ($\geq 1\%$), rhinorrhea (1%)

<1%:

Cardiovascular: Bradycardia, circulatory shock, syncope

Dermatologic: Diaphoresis, pruritus, urticaria

Gastrointestinal: Anorexia, vomiting

Genitourinary: Urinary tract infection

Hematologic & oncologic: Anemia, lymphadenopathy, neutropenia

Infection: Infection

Nervous system: Tonic-clonic type of status epilepticus

Neuromuscular & skeletal: Hyperkinetic muscle activity

Ophthalmic: Mydriasis, nystagmus disorder

Respiratory: Increased cough

Frequency not defined:

Cardiovascular: ECG changes, venous thrombosis

Endocrine & metabolic: Change in libido

Gastrointestinal: Altered salivation, constipation, gastrointestinal distress, nausea

Genitourinary: Urinary incontinence, urinary retention

Hepatic: Increased serum alkaline phosphatase, increased serum transaminases, jaundice

Local: Localized phlebitis

Nervous system: Anterograde amnesia, central nervous system depression, depression, drug abuse, drug dependence, drug withdrawal, fatigue, hypoactivity, lethargy, myasthenia, paradoxical central nervous system stimulation, psychiatric signs and symptoms, rebound anxiety, slurred speech, tremor

Ophthalmic: Blurred vision, diplopia

Respiratory: Hypoventilation

Postmarketing: Miscellaneous: Paradoxical reaction

Contraindications

Hypersensitivity to diazepam or any component of the formulation; acute narrow-angle glaucoma.

Injection: Additional contraindications: Untreated open-angle glaucoma.

Oral: Additional contraindications: Untreated open-angle glaucoma; use in infants <6 months of age, myasthenia gravis, severe respiratory impairment, severe hepatic impairment, sleep apnea syndrome.

Warnings/Precautions

Concerns related to adverse effects:

- Anterograde amnesia: Benzodiazepines have been associated with anterograde amnesia (Nelson 1999).
- CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery, driving).
- Paradoxical reactions: Paradoxical reactions, including hyperactive or aggressive behavior, have been reported with benzodiazepines; risk may be increased in adolescent/pediatric, geriatric patients, or patients with a history of alcohol use disorder or psychiatric/personality disorders (Mancuso 2004).
- Sleep-related activities: Hazardous sleep-related activities such as sleep-driving, cooking and eating food, and making phone calls while asleep have been noted with benzodiazepines (Dolder 2008).
- Suicidal ideation: Intranasal: Pooled analysis of trials involving various antiseizure medications (regardless of indication) showed an increased risk of suicidal thoughts/behavior (incidence rate: 0.43% of treated patients compared to 0.24% of patients receiving placebo); risk observed as early as 1 week after initiation and continued through duration of trials (most trials ≤ 24 weeks). Monitor all patients for notable changes in behavior that might indicate suicidal thoughts or depression; notify health care provider immediately if symptoms occur.

Disease-related concerns:

- Bariatric surgery: Presurgical assessment of the indication for use, symptoms, and goals of therapy should be documented to enable postsurgical assessment. Monitor for continued efficacy after bariatric surgery and consider switching to an alternate medication if symptoms worsen or adverse reactions occur.
- Convulsive disorders: When used as an adjunct in treating convulsive disorders, an increase in frequency/severity of tonic-clonic seizures may occur and require dose adjustment of antiseizure medication. Abrupt withdrawal may result in a temporary increase in the frequency and/or severity of seizures.
- Depression: Avoid use in patients with depression because of concerns about worsening mood symptoms, particularly if suicidal risk may be present, except for acute or emergency situations (eg, acute agitation, status epilepticus) (Stein 2025).
- Glaucoma: May be used in patients with open-angle glaucoma who are receiving appropriate therapy.
- Hepatic impairment: Use with caution in patients with hepatic impairment.
- Renal impairment: Use with caution in patients with renal impairment.
- Respiratory disease: Reduce dose or avoid use in patients with respiratory disease, including chronic obstructive pulmonary disease or sleep apnea. Benzodiazepines may cause significant respiratory depression.

Special populations:

- Debilitated patients: Use with caution; active metabolites with extended half-lives may lead to delayed accumulation and adverse effects; limit dose to smallest effective amount and increase gradually and as tolerated to avoid adverse reactions.
- Older adult patients: Use with caution; active metabolites with extended half-lives may lead to delayed accumulation and adverse effects; limit dose to smallest effective amount and increase gradually and as tolerated to avoid adverse reactions. Older adults may be at an increased risk of death with use; risk has been found highest within the first 4 months of use in older adult dementia patients (Jennum 2015; Saarelainen 2018).
- Fall risk: Use with extreme caution in patients who are at risk of falls; benzodiazepines have been associated with falls and traumatic injury (Nelson 1999).
- Obese patients: Use benzodiazepines with caution in obese patients; may have prolonged action when discontinued.
- Psychotic patients: Use of diazepam is not recommended in place of appropriate therapy.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol and/or sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol and/or benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.
- Parenteral: Vesicant; ensure proper needle or catheter placement prior to and during administration; avoid extravasation. Acute hypotension, muscle weakness, apnea, and/or cardiac arrest have occurred with parenteral administration. Acute effects may be more prevalent in patients receiving concurrent barbiturates, opioids, or ethanol. Appropriate resuscitative equipment and qualified personnel should be available during administration and monitoring. Avoid use of the injection in patients in shock, coma, or in acute ethanol intoxication with depression of vital signs. Intra-arterial injection should be avoided. Tonic status epilepticus has been precipitated in patients treated with diazepam IV for absence status or absence variant status.
- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP ["Inactive" 1997]; Wilson 2000; Wilson 2005; Zar 2007).
- Rectal gel: Administration of rectal gel should only be performed by individuals trained to recognize characteristic seizure activity for which the product is indicated, and capable of monitoring response to determine need for additional medical intervention. Not recommended for chronic, daily use. Use with caution in patients with neurologic damage.

Other warnings/precautions:

- Abuse, misuse, and substance use disorder: Counsel patients at increased risk on proper use and monitoring for signs and symptoms of abuse, misuse, and substance use disorder. Institute early treatment or refer patients in whom substance use disorder is suspected. Limit dosages and durations to the minimum required.
- Appropriate use: Does not have analgesic, antidepressant, or antipsychotic properties.
- Dependence and withdrawal reactions: The continued use of benzodiazepines, including diazepam, may lead to clinically significant physical dependence, and abrupt discontinuation or rapid dose reduction may precipitate acute withdrawal reactions, which may be life-threatening. Some patients may develop a protracted withdrawal syndrome lasting >12 months; may be difficult to differentiate withdrawal symptoms from reemergence or continuation of symptoms for which benzodiazepines

were prescribed. Flumazenil may cause withdrawal in patients receiving long-term benzodiazepine therapy.

- **Tolerance:** Diazepam is a long half-life benzodiazepine. Duration of action after a single dose is determined by redistribution rather than metabolism. Tolerance develops to the sedative, hypnotic, and antiseizure effects. It does not develop to the anxiolytic or skeletal muscle relaxing effects (Vinkers 2012). Chronic use of this agent may increase the perioperative benzodiazepine dose needed to achieve desired effect.

Warnings: Additional Pediatric Considerations

Neonates and young infants have decreased metabolism of diazepam and desmethyldiazepam (active metabolite), both can accumulate with repeated use and cause increased toxicity.

Some dosage forms may contain propylene glycol; in neonates, large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009). Diazepam Intensol Oral Solution (Concentration) contains 19% alcohol.

Product Availability

Libervant buccal film has been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Concentrate, Oral:

diazePAM Intensol: 5 mg/mL (30 mL) [contains alcohol, usp; unflavored flavor]

Generic: 5 mg/mL (30 mL)

Film, Buccal:

Libervant: 5 mg (1 ea [DSC], 2 ea [DSC]); 7.5 mg (1 ea [DSC], 2 ea [DSC]) [contains edetate (edta) disodium]

Libervant: 10 mg (1 ea [DSC], 2 ea [DSC]) [contains disodium edta]

Libervant: 12.5 mg (1 ea [DSC], 2 ea [DSC]); 15 mg (1 ea [DSC], 2 ea [DSC]) [contains edetate (edta) disodium]

Gel, Rectal:

Diastat AcuDial: 10 mg (1 ea [DSC]); 20 mg (1 ea [DSC]) [contains alcohol, usp, benzoic acid, sodium benzoate]

Diastat Pediatric: 2.5 mg (1 ea [DSC]) [contains benzoic acid, benzyl alcohol, propylene glycol, sodium benzoate]

Generic: 2.5 mg (1 ea); 10 mg (1 ea); 20 mg (1 ea)

Liquid, Nasal:

Valtoco 10 MG Dose: 10 mg/0.1 mL (1 ea) [contains alcohol, usp]

Valtoco 5 MG Dose: 5 mg/0.1 mL (1 ea) [contains alcohol, usp]

Liquid Therapy Pack, Nasal:

Valtoco 15 MG Dose: 2 devices, 7.5 mg/0.1 mL each (1 ea) [contains alcohol, usp]

Valtoco 20 MG Dose: 2 devices, 10 mg/0.1 mL each (1 ea) [contains alcohol, usp]

Solution, Injection:

Generic: 5 mg/mL (2 mL, 10 mL); 10 mg/2 mL (2 mL)

Solution, Oral:

Generic: 5 mg/5 mL (5 mL, 473 mL, 500 mL)

Solution Auto-injector, Intramuscular:

Generic: 10 mg/2 mL (2 mL [DSC])

Tablet, Oral:

Valium: 2 mg [scored; contains corn starch]

Valium: 5 mg [scored; contains corn starch, fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Valium: 10 mg [scored; contains corn starch, fd&c blue #1 (brilliant blue)]

Generic: 2 mg, 5 mg, 10 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Concentrate (diazepam Intensol Oral)

5 mg/mL (per mL): \$1.43

Concentrate (diazepam Oral)

5 mg/mL (per mL): \$5.00

Gel (diazepam Rectal)

2.5 mg (per each): \$306.90

10 mg (per each): \$364.06

20 mg (per each): \$364.06

Liquid (Valtoco 10 MG Dose Nasal)

10MG/0.1ML (per each): \$458.88

Liquid (Valtoco 5 MG Dose Nasal)

5MG/0.1ML (per each): \$458.88

Liquid Therapy Pack (Valtoco 15 MG Dose Nasal)

2 x 7.5MG/0.1ML (per each): \$458.88

Liquid Therapy Pack (Valtoco 20 MG Dose Nasal)

2 x 10MG/0.1ML (per each): \$458.88

Solution (diazepam Injection)

5 mg/mL (per mL): \$11.18 - \$24.00

Tablets (diazepam Oral)

2 mg (per each): \$0.10 - \$0.24

5 mg (per each): \$0.16 - \$0.32

10 mg (per each): \$0.31 - \$0.43

Tablets (Valium Oral)

2 mg (per each): \$9.62

5 mg (per each): \$14.97

10 mg (per each): \$25.21

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Gel, Rectal:

Diastat: 10 mg (2 mL) [contains alcohol, usp, benzoic acid, benzyl alcohol, propylene glycol, sodium benzoate]

Solution, Injection:

Generic: 5 mg/mL (2 mL)

Tablet, Oral:

Valium: 5 mg [contains corn starch]

Generic: 2 mg, 5 mg, 10 mg

Controlled Substance

C-IV

Administration: Adult

Oral: Administer with food or water. Dilute or mix oral concentrate with water, juice, soda, apple-sauce, or pudding before use; measure dose only with calibrated dropper provided.

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN; Lucas E. Orth, PharmD, BCPPS; Russel J. Roberts, PharmD, BCCCP, FCCM.

Oral concentrate solution (5 mg/mL, commercially available): Enteral feeding tube administration utilizing diazepam oral concentrate solution (5 mg/mL) is not recommended; diazepam liquid may bind to enteral feeding tube material (Ref).

Oral solution (1 mg/mL, commercially available):

Note: Enteral feeding tube administration utilizing diazepam oral solution is not preferred; diazepam liquid may bind to enteral feeding tube material (Ref). When alternatives are not available and use of oral solution is deemed necessary for feeding tube administration, some institutions have successfully administered oral solution; consider the risks versus benefits and monitor closely for therapeutic effect (Ref).

Gastric (eg, NG, G-tube) tubes: Dilute dose with at least an equivalent volume of purified water immediately prior to administration to reduce osmolality and viscosity. Draw up solution into enteral dosing syringe and administer via feeding tube (Ref).

General guidance: Hold enteral nutrition (EN) during diazepam administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (ASPEN [Boullata 2017]). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Oral tablet:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes (≥ 8 French): Crush tablet(s) into a fine powder and disperse in 10 to 20 mL purified water immediately prior to administration; dispersion may take up to 5 minutes. Draw up mixture into enteral dosing syringe and administer via feeding tube (Ref). Alternatively, may place intact tablet into the barrel of an empty enteral dosing syringe or cup with 10 to 20 mL of purified water and allow to dissolve over 2 to 5 minutes; if necessary, shake syringe or swirl cup and then immediately administer via feeding tube (Ref).

General guidance: Hold EN during diazepam administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse container used for preparation with purified water (eg, 10 mL); draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Intranasal: Do not test or prime before use. Administer one spray into one nostril. Some doses require an additional spray into the alternate nostril; refer to dosing for additional details. Do not administer a second dose if the patient is having trouble breathing or is excessively sedated.

IM: Administer (undiluted) deep into the muscle mass.

IV: Administer undiluted by slow IV push; do not mix with other solutions or medications. Rapid injection may cause respiratory depression or hypotension. In adults, maximum infusion rate is 5 mg/minute. Do not administer through small veins (eg, dorsum of hand/wrist). Avoid intra-arterial administration. Continuous infusion is not recommended because of precipitation in IV fluids and absorption of drug into infusion bags and tubing.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm or cold compresses; initiate hyaluronidase antidote (if indicated) (Ref).

Hyaluronidase: Intradermal or SUBQ: Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if infiltrated catheter remains in place, administer through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution (Ref).

Rectal gel: Prior to administration, confirm that prescribed dose is visible and correct and that the green "ready" band is visible. Place patient on side (facing person responsible for monitoring), with top leg bent forward. Insert rectal tip (lubricated) gently into rectum until rim fits snugly against rectal opening; push plunger gently over 3 seconds. After additional 3 seconds, remove syringe; hold buttocks together while slowly counting to 3 to prevent leakage; keep patient on side, facing towards you, and continue to observe patient; discard any unused medication, syringe, and all used materials; do not reuse; see manufacturer's Administration and Disposal Instructions. May consider use of parenteral formulation rectally if gel not available (Ref).

Administration: Pediatric

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh,

BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN; Lucas E. Orth, PharmD, BCPPS; Russel J. Roberts, PharmD, BCCCP, FCCM.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral : Administer with food or water.

Oral concentrate solution (5 mg/mL) (commercially available): Dilute or mix dose with water, juice (except grapefruit juice), soda, applesauce, or pudding before use. Measure dose only with calibrated dropper provided.

Administration via feeding tube: NOT recommended: Enteral feeding tube administration utilizing diazepam oral concentrate solution (5 mg/mL) is not recommended; diazepam liquid may bind to enteral feeding tube material (Ref).

Oral solution (1 mg/mL) (commercially available): Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Administration via feeding tube: Note: Enteral feeding tube administration utilizing diazepam oral solution is not preferred; diazepam liquid may bind to enteral feeding tube material (Ref). When alternatives are not available and use of oral solution is deemed necessary for feeding tube administration, some institutions have successfully administered oral solution; consider the risks versus benefits and monitor closely for therapeutic effect (Ref).

Gastric (eg, NG, G-tube) tubes: Dilute dose with at least an equivalent volume of purified water immediately prior to administration to reduce osmolality and viscosity. Draw up solution into enteral dosing syringe and administer via feeding tube (Ref).

General guidance: Hold enteral nutrition during diazepam administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Tablet:

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes (≥ 8 French): Crush tablet(s) into a fine powder and disperse in 10 to 20 mL purified water immediately prior to administration; dispersion may take up to 5 minutes. Draw up mixture into enteral dosing syringe and administer via feeding tube (Ref). Alternatively, may place intact tablet into the barrel of an empty enteral dosing syringe or cup with 10 to 20 mL of purified water and allow to dissolve over 2 to 5 minutes; if necessary, shake syringe and then immediately administer via feeding tube (Ref).

General guidance: Hold enteral nutrition during diazepam administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Buccal: Film: Libervant: Keep in foil pouch until ready for use. With clean and dry hands, following directions provided on foil, open pouch and remove green-colored film. Stretch either cheek open with one hand and place the film on the inside of that cheek with other hand, then remove fingers from patient's mouth. Film will stick to inside of patient's cheek and dissolve; mouth may be opened or closed. Do not rub the film into the cheek or place film on the teeth. If film is spit out or blown out immediately after placement, attempt to place new film; if unable, alert emergency personnel.

If film is accidentally swallowed or chewed during administration, no replacement dose is necessary. Do not administer with liquids.

Intranasal: Nasal solution: Valtoco: Device comes ready to use; do not test or prime device before use. Each device delivers 1 spray only. Administer 1 spray into 1 nostril from a single device. If a second device is needed for the full dose, administer the second spray in the alternate nostril from a new device. A second dose should not be administered if the patient is having trouble breathing or excessive sedation.

Parenteral:

IM: Administer undiluted (5 mg/mL) deep into muscle mass.

IV: Administer undiluted (5 mg/mL) direct IV; do not mix with other solutions or medications. Rapid injection may cause respiratory depression or hypotension; infants and children: Do not exceed 1 to 2 mg/minute IV push; adults: 5 mg/minute. Continuous infusion is not recommended because of precipitation in IV fluids, absorption of drug into infusion bags and tubing, and propylene glycol exposure.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop IV administration immediately; leave cannula/needle in place, but do NOT flush the line; gently aspirate extravasated solution; remove needle/cannula; elevate extremity. Apply dry cold or warm compresses (Ref).

Rectal:

Gel: Prescribed dose must be "dialed in" and locked before dispensing; consult package insert for directions on setting prescribed dose. Prior to administration, confirm that the prescribed dose is visible and correct and that the green "ready" band is visible.

Diastat AcuDial and Diastat: Place patient on side (facing person responsible for monitoring), with top leg bent forward. Insert rectal tip (lubricated) gently into rectum until rim fits snug against rectal opening; push plunger gently over 3 seconds. After additional 3 seconds, remove syringe; hold buttocks together while slowly counting to 3 to prevent leakage; keep patient on side, facing towards you and continue to observe patient; discard any unused medication, syringe, and all used materials safely away from children; do not reuse; see Administration and Disposal Instructions that come with product.

Injectable solution: If withdrawing dose from an ampule, use a filter needle. Administer as undiluted 5 mg/mL solution.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Buccal film: Store at 20°C to 25°C (68°F to 77°F); excursions permitted from 15°C to 30°C (59°F to 86°F).

Injection: Store at 20°C to 25°C (68°F to 77°F). Protect from light. Do not refrigerate autoinjector.

Nasal spray: Store at 20°C to 25°C (68°F to 77°F); excursions permitted from 15°C to 30°C (59°F to 86°F). Do not freeze. Protect from light. Only open blister pack immediately prior to administration.

Oral solution: Store at 20°C to 25°C (68°F to 77°F). Protect from light. Discard opened bottle of concentrated oral solution after 90 days.

Rectal gel: Store at 25°C (77°F); excursion permitted to 15°C to 30°C (59°F to 86°F).

Tablet: Store at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Alcohol withdrawal syndrome: Symptomatic relief of acute agitation, tremor, impending or acute delirium, delirium tremens, and hallucinosis associated with alcohol withdrawal.

Anxiety, acute/severe: Short-term relief of severe anxiety symptoms.

Anxiety disorders: Management of anxiety disorders.

Muscle spasm, spasticity, and/or rigidity: As an adjunct for the relief of skeletal muscle spasm due to reflex spasm caused by local pathology (eg, inflammation of muscles or joints, secondary to trauma); spasticity caused by upper motor neuron disorders (eg, cerebral palsy, paraplegia); athetosis; stiff-man syndrome; and tetanus.

Procedural anxiety, premedication: Relief of anxiety and tension in patients undergoing surgical procedures; prior to cardioversion for the relief of anxiety and tension and to diminish patient's recall; as an adjunct prior to endoscopic procedures for apprehension, anxiety, or acute stress reactions and to diminish patient's recall.

Note: Use of diazepam in patients undergoing cardioversion or endoscopic procedures has been superseded by agents with a more pharmacokinetically favorable profile (eg, midazolam) (Thomas 2014; Triantafillidis 2013).

Seizures, acute, active: As an adjunct in the treatment of intermittent, stereotypic episodes of frequent seizure activity (ie, seizure clusters, acute repetitive seizures) that are distinct from a patient's usual seizure pattern in patients with epilepsy; adjunct in severe recurrent convulsive seizures.

Status epilepticus: Adjunct in status epilepticus.

Use: Off-Label: Adult

Hydroxychloroquine/chloroquine toxicity (severe); Intoxication (cocaine, methamphetamine, and other sympathomimetics); Neuroleptic malignant syndrome; Opioid withdrawal (autonomic instability and agitation) in the emergency setting; Organophosphate poisoning; Serotonin syndrome (serotonin toxicity); Vertigo, acute episodes

Medication Safety Issues

Sound-alike/look-alike issues:

DiazePAM may be confused with diazoxide, diLTIAZem, Ditropan, LORazepam

Valium may be confused with Valcyte

Older Adult: High-Risk Medication:

Beers Criteria: Diazepam is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older due to risk of abuse, misuse, physical dependence, and addiction. In addition, older adults have an increased risk of impaired cognition, delirium, falls, fractures, and motor vehicle accidents with benzodiazepine use, and slower metabolism of long-acting benzodiazepines (eg, diazepam). However, benzodiazepines may be appropriate in the elderly when used for seizure disorders, rapid eye movement sleep behavior disorder, benzodiazepine or ethanol withdrawal, severe generalized anxiety disorder, or periprocedural anesthesia (Beers Criteria [AGS 2023]).

Diazepam is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) with a history of recurrent falls due to an increased risk of falls. Use of benzodiazepines is not recommended for ≥ 4 weeks; limit insomnia treatment to < 2 weeks. In addition, some disease states of concern include dementia and respiratory failure (acute or chronic) (O'Mahony 2023).

Metabolism/Transport Effects

Substrate of CYP1A2 (Minor), CYP2B6 (Minor), CYP2C19 (Major), CYP2C9 (Minor), CYP3A4 (Major);

Note: Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ajmaline: DiazePAM may increase adverse/toxic effects of Ajmaline. Specifically, the risk for cholestasis may be increased. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ARIPiprazole Lauroxil: May increase CNS depressant effects of Benzodiazepines. ARIPiprazole Lauroxil may increase hypotensive effects of Benzodiazepines. Specifically, the risk of orthostatic hypotension may be increased. *Risk C: Monitor*

ARIPiprazole: May increase CNS depressant effects of Benzodiazepines. ARIPiprazole may increase hypotensive effects of Benzodiazepines. Specifically, orthostatic hypotension may be increased. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benzodiazepines: CNS Depressants may increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Beta-Acetyldigoxin: Benzodiazepines may increase serum concentrations of Beta-Acetyldigoxin. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bucizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cannabinoid-Containing Products: CNS Depressants may increase CNS depressant effects of Cannabinoid-Containing Products. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Certoparin: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: Benzodiazepines may increase adverse/toxic effects of CloZAPine. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Corticosteroids (Orally Inhaled): Benzodiazepines may increase adverse/toxic effects of Corticosteroids (Orally Inhaled). Specifically, the risk of pneumonia may be increased. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

CYP2C19 Inducers (Moderate): May decrease serum concentrations of DiazePAM. *Risk C: Monitor*

CYP2C19 Inhibitors (Moderate): May increase serum concentrations of DiazePAM. *Risk C: Monitor*

CYP2C19 Inhibitors (Strong): May increase serum concentrations of DiazePAM. *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of DiazePAM. *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of DiazePAM. *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of DiazePAM. *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of DiazePAM. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimenhydrinate: CNS depressant effects of CNS Depressants and Dimenhydrinate may increase with coadministration. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Disulfiram: May increase adverse/toxic effects of Products Containing Ethanol. Management: Do not use disulfiram with dosage forms that contain ethanol. *Risk X: Avoid*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Fosphenytoin-Phenytoin: May decrease serum concentrations of DiazePAM. DiazePAM may increase serum concentrations of Fosphenytoin-Phenytoin. *Risk C: Monitor*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of DiazePAM. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

HydroXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Ilaprazole: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumacaftor and Ivacaftor: May decrease serum concentrations of CYP2C19 Substrates (High risk with Inducers). *Risk C: Monitor*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melatonin: May increase sedative effects of Benzodiazepines. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methadone: Benzodiazepines may increase CNS depressant effects of Methadone. Management: Clinicians should generally avoid concurrent use of methadone and benzodiazepines when possible; any combined use should be undertaken with extra caution. *Risk D: Consider Therapy Modification*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Products Containing Ethanol. Management: Avoid products containing alcohol in patients treated with methotrimeprazine. *Risk X: Avoid*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of DiazePAM. Metoclopramide may increase serum concentrations of DiazePAM. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: Benzodiazepines may increase CNS depressant effects of Mirtazapine. *Risk X: Avoid*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nirmatrelvir and Ritonavir: May increase serum concentrations of DiazePAM. Nirmatrelvir and Ritonavir may decrease serum concentrations of DiazePAM. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANZapine: Benzodiazepines may increase adverse/toxic effects of OLANZapine. Management: Monitor closely for hypotension, respiratory or central nervous system depression, and bradycardia if olanzapine is combined with benzodiazepines. Use of parenteral benzodiazepines with IM olanzapine is not recommended. *Risk C: Monitor*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ombitasvir, Paritaprevir, and Ritonavir: May decrease serum concentrations of DiazePAM. Ombitasvir, Paritaprevir, and Ritonavir may decrease active metabolite exposure of DiazePAM. *Risk C: Monitor*

Ombitasvir, Paritaprevir, Ritonavir, and Dasabuvir: May decrease serum concentrations of DiazePAM. Ombitasvir, Paritaprevir, Ritonavir, and Dasabuvir may decrease active metabolite exposure of DiazePAM. *Risk C: Monitor*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): Benzodiazepines may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). *Risk X: Avoid*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procabazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QUetiapine: CNS Depressants may increase CNS depressant effects of QUetiapine. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Risperidone: CNS depressant effects of CNS Depressants and Risperidone may increase with coadministration. *Risk C: Monitor*

Ritonavir: May increase serum concentrations of Diazepam. Ritonavir may decrease serum concentrations of Diazepam. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Secnidazole: Products Containing Ethanol may increase adverse/toxic effects of Secnidazole. *Risk X: Avoid*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Teduglutide: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Theophylline Derivatives: May decrease therapeutic effects of Benzodiazepines. *Risk C: Monitor*

Thioridazine: CNS Depressants may increase CNS depressant effects of Thioridazine. *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

Tiagabine: CNS Depressants may increase CNS depressant effects of Tiagabine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: May increase CNS depressant effects of DiazePAM. Valproic Acid and Derivatives may increase serum concentrations of DiazePAM. DiazePAM may decrease serum concentrations of Valproic Acid and Derivatives. Valproic Acid and Derivatives may decrease serum concentrations of DiazePAM. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Yohimbine: May decrease therapeutic effects of Antianxiety Agents. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Evaluate pregnancy status prior to use. Pregnancy testing is recommended before treating acute alcohol withdrawal symptoms; however, treatment should not be delayed in symptomatic patients if pregnancy status is unknown (ASAM 2020).

Diazepam is not a first-line medication to treat anxiety disorders prior to conception in patients who are treatment naive or who do not have a history of effective treatment with a specific medication. Patients effectively treated may continue their current medication when planning a pregnancy unless contraindications exist. Management of mental health conditions in patients who could become pregnant should be based on a shared decision-making process that considers the possibility of pregnancy during treatment (ACOG 2023; BAP [McAllister-Williams 2017]).

Manage epilepsy in patients who could become pregnant based on a shared decision-making process that optimizes seizure control and considers the possibility of pregnancy during treatment (Pack 2024). Regularly discuss age-specific and developmental needs, including pregnancy planning and contraceptive options during the patient's reproductive lifespan (ACOG 2020; NICE 2022).

Pregnancy Considerations

Diazepam and its metabolites (N-desmethyldiazepam, temazepam, and oxazepam) cross the placenta (Kanto 1982; McElhatton 1994).

In-utero exposure to benzodiazepines has the potential to cause harm to the fetus. Teratogenic effects have been observed in some studies; however, a clear association has not been reported and additional data are needed (Bellantuono 2013; Chuang 2024; Freeman 2018; Grigoriadis 2019; Tinker 2019; Wu 2024). Exposure to a benzodiazepine late in pregnancy may cause neonatal sedation (hypotonia, lethargy, respiratory depression) and/or symptoms of neonatal withdrawal (feeding difficulties, hyperreflexia, inconsolable crying, irritability, restlessness, tremors). Monitor newborns exposed to diazepam in utero for adverse events. Data related to long-term effects on neurodevelopment following maternal use of benzodiazepines are inconclusive (Andrade 2024; Radojčić 2017; Sundbakk 2025; Wang 2022).

Benzodiazepines are one of the preferred options for treating pregnant patients with alcohol withdrawal. Inpatient treatment is recommended during pregnancy. Agents with a shorter onset and duration of action should be used if near delivery. Monitor newborns for fetal alcohol spectrum disorder, sedative withdrawal, and intoxication (ASAM 2020).

Epilepsy is associated with adverse maternal and fetal outcomes (Kuang 2024; Mazzone 2023). Convulsive seizures should be minimized to reduce risks to the fetus and pregnant patient. Use caution if removing or replacing an effective seizure medication in patients who become pregnant during therapy. Folic acid supplementation prior to and during pregnancy minimizes the risk of congenital malformations and poor neurodevelopment (Pack 2024). Treatment of status epilepticus during pregnancy should consider gestational age and etiology of seizures (eg, eclampsia versus other causes) (NCS [Brophy 2012]; Rajiv 2019).

Untreated and undertreated mental health conditions are associated with adverse pregnancy outcomes. Anxiety disorders during pregnancy are associated with low birth weight, preterm birth, and adverse behavioral outcomes in the offspring. Discontinuing effective medications during pregnancy increases the risk of relapse. Management should be made as part of a shared decision-making process. Agents other than diazepam may be preferred when treatment for anxiety is initiated for the first time during pregnancy. The use of benzodiazepines for the treatment of perinatal anxiety should be avoided or used sparingly until preferred therapy is initiated and effective, or when preferred therapy is not effective. When medications are used, the lowest effective dose of a single agent is recommended. Optimize dosing prior to changing a medication or adding additional agents whenever possible. Monthly monitoring for symptom improvement with a validated screening tool during pregnancy is recommended. Manage side effects as needed (ACOG 2023).

Data collection to monitor pregnancy outcomes following exposure to diazepam is ongoing. Encourage patients to enroll in the North American Antiepileptic Drug Pregnancy Registry (1-888-233-2334) or the National Pregnancy Registry for Psychiatric Medications (1-866-961-2388).

Breastfeeding Considerations

Diazepam and its metabolites are present in breast milk.

Data related to the presence of diazepam in breast milk are available from multiple sources:

- The presence of diazepam was evaluated in 4 patients who received diazepam 20 mg IV prior to delivery, then diazepam 10 mg orally once daily from days 3 to 6 postpartum. Breast milk was sampled ~9 and ~23 hours after the oral doses. Diazepam and methyldiazepam were present in breast milk in concentrations ranging from 12 to 43 ng/mL and 12 to 85 ng/mL, respectively (Brandt 1976).
- Breast milk and infant serum were sampled following administration of diazepam 10 mg 3 times daily to 3 mothers for 6 days after delivery. Diazepam breast milk concentrations were 51 ng/mL (4 days postpartum) and 78 ng/mL (6 days postpartum). Methyldiazepam breast milk concentrations were 28 ng/mL (4 days postpartum) and 52 ng/mL (6 days postpartum). Diazepam and methyldiazepam were present in the breastfed infant plasma (Erkkola 1972).
- Additional studies have confirmed the presence of diazepam and methyldiazepam in breast milk; however, the maternal doses associated with reported breast milk concentrations were not clearly stated (Cole 1975; Dusci 1990; Wesson 1985).
- The oxazepam metabolite has been detected in the urine of an infant exposed to diazepam via breast milk (Patrick 1972).
- Relative infant doses (RID) of up to 11% have been reported for diazepam and methyldiazepam (McElhatton 1994). In general, breastfeeding is considered acceptable when the RID of a medication is <10% (Anderson 2016; Ito 2000); however, some sources note breastfeeding should only be considered if the RID is <5% for psychotropic agents (Eleftheriou 2024).

Drowsiness, lethargy, or weight loss in breastfed infants have been observed in case reports following maternal use of some benzodiazepines (Iqbal 2002), including diazepam (Patrick 1972; Wesson 1985). Clinically significant weight loss or jaundice were observed in infants of mothers who received IV diazepam at doses >0.1 mg/kg as an anesthetic for tubal ligation. Oral maternal doses of diazepam 5 mg administered prior to the procedure and lower IV doses administered during the procedure were not associated with these effects. Patients were between 1 and 6 days postpartum, and most infants were fully breastfed (Gilder 2021).

Diazepam has a long half-life that varies by age and is longer in preterm infants and full-term newborns, likely due to incomplete maturation of the metabolic pathway. Diazepam may accumulate in the breastfed infant, especially preterm infants or those exposed to chronic maternal doses (Davanzo 2013). Significant accumulation may occur even if the maternal dose is low (Wesson 1985).

Due to the potential for serious adverse events in a breastfed infant (eg, sedation, withdrawal symptoms) breastfeeding during diazepam therapy is not recommended by the manufacturer. Avoid benzodiazepines with a long half-life or multiple active metabolites (such as diazepam) in patients who are breastfeeding. A benzodiazepine with a short half-life can be considered for the treatment of anxiety in patients who are breastfeeding (Eleftheriou 2024).

Monitoring Parameters

Heart rate, respiratory rate, blood pressure, and mental status; liver enzymes and CBC with long-term therapy; clinical signs of propylene glycol toxicity (for continuous high-dose and/or long duration intravenous use) including serum creatinine, BUN, serum lactate, osmol gap. **Note:** An osmol gap of ≥ 10 was predictive of elevated propylene glycol concentrations; values of ≥ 12 suggest propylene glycol toxicity (Arroliga 2004; Barnes 2006; Yahwak 2008).

Mechanism of Action

Long-acting benzodiazepine. Binds to stereospecific benzodiazepine receptors on the postsynaptic GABA neuron at several sites within the central nervous system, including the limbic system, reticular formation. Enhancement of the inhibitory effect of GABA on neuronal excitability results by increased neuronal membrane permeability to chloride ions. This shift in chloride ions results in hyperpolarization (a less excitable state) and stabilization. Benzodiazepine receptors and effects appear to be linked to the GABA-A receptors. Benzodiazepines do not bind to GABA-B receptors.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Sedation: Pediatric patients: IV: 4 to 5 minutes (Krauss 2006).

Status epilepticus: IV: 1 to 3 minutes; Rectal: 2 to 10 minutes.

Duration of action:

Classified as a long-acting benzodiazepine; classification based on benzodiazepines with half-life >40 hours (Griffin 2013).

Indication-specific durations:

Sedation: Pediatric patients: IV: 60 to 120 minutes (Krauss 2006).

Absorption:

Oral: Well absorbed (>90%); delayed and decreased when administered with a moderate fat meal.

Rectal: Well absorbed.

Distribution: V_d :

Buccal film: 1.46 L/kg.

Intranasal: 0.8 to 1 L/kg.

IV: 1.2 L/kg (range: 0.6 to 2 L/kg) (Greenblatt 1989a).

Oral: 1.1 L/kg (range: 0.6 to 1.8 L/kg) (Greenblatt 1989b).

Rectal: 1 L/kg.

Protein binding:

Buccal film: 95% to 98%.

Intranasal: 95% to 98%.

Oral: Neonates: 84% to 86% (Milsap 1994; Morselli 1980); Adults: 98%.

Rectal: 95% to 98%.

Metabolism: Hepatic; diazepam is N-demethylated by CYP3A4 and 2C19 to the active metabolite N-desmethyldiazepam, and is hydroxylated by CYP3A4 to the active metabolite temazepam. N-desmethyldiazepam and temazepam are both further metabolized to oxazepam. Temazepam and oxazepam are largely eliminated by glucuronidation.

Bioavailability:

IM: >90% (Lamson 2011).

Intranasal: 97%.

Oral: >90%.

Rectal: 90%.

Half-life elimination: **Note:** Diazepam accumulates upon multiple dosing and the terminal elimination half-life is slightly prolonged.

Buccal film: Parent: ~86 hours; Desmethyldiazepam: ~147 hours.

IM:

Premature neonates (GA: 28 to 34 weeks): 54 hours.

Infants: ~30 hours (Morselli 1973).

Children 3 to 8 years: 18 hours (Morselli 1973).

Adults: Parent: ~60 to 72 hours; Desmethyldiazepam: ~152 to 174 hours (Lamson 2011).

Intranasal: ~49 hours.

IV: Parent: 33 to 45 hours; Desmethyldiazepam: 87 hours (Cloyd 1998; Greenblatt 1989a).

Oral: Parent: 44 to 48 hours; Desmethyldiazepam: 100 hours (Greenblatt 1989b).

Rectal: Parent: 45 to 46 hours; Desmethyldiazepam: 71 to 99 hours (Cloyd 1998).

Time to peak:

Buccal film: ~1 hour.

IM: Median: 1 hour (range: 0.25 to 2 hours) (Lamson 2011).

Intranasal: ~1.5 hours.

IV: ~1 minute (Cloyd 1998).

Oral: 15 minutes to 2.5 hours (1.25 hours when fasting; 2.5 hours with food) (Greenblatt 1989b).

Rectal: 1.5 hours.

Excretion: Urine (predominantly as glucuronide conjugates).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: In mild and moderate cirrhosis, the average half-life increases 2- to 5-fold, with individual half-lives over 500 hours reported. There is also an increase in volume of distribution, and average clearance decreases by almost half. Half-life is also prolonged with hepatic fibrosis to 90 hours (range: 66 to 104 hours), with chronic active hepatitis to 60 hours (range: 26 to 76 hours), and with acute viral hepatitis to 74 hours (range: 49 to 129 hours). In chronic active hepatitis, clearance is decreased by almost half.

Older adult: The half-life is increased by approximately 1 hour for each year of age beginning with a half-life of 20 hours at 20 years of age, as the volume of distribution is increased, and clearance is decreased. Consequently, there may be lower peak concentrations, higher trough concentration with multiple doses, and it may take longer to reach steady state.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Valium: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/013263Orig1s100lbl.pdf

Valtoco nasal spray: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/211635s008lbl.pdf#page=24

Patient Counseling Points

(For additional information see "Diazepam: Patient drug information")

What is this drug used for?

- It is used to relax muscles.
- It is used to treat alcohol withdrawal.
- It is used to treat anxiety.
- It is used to help control certain kinds of seizures.
- It may be given for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Feeling sleepy
- Feeling tired or weak
- Muscle weakness

Nose spray and rectal gel:

- Headache

Nose spray:

- Nose irritation

Injection:

- Irritation where the shot is given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Shortness of breath
- Change in balance
- Feeling confused
- Hallucinations (seeing or hearing things that are not there)
- Memory problems or loss
- Very bad dizziness or passing out
- Muscle spasm
- Twitching
- Trouble sleeping
- Change in eyesight
- Slurred speech
- Upset stomach, throwing up, or loss of appetite that is severe or does not go away
- Depression like feeling nervous, restless, or grouchy; panic attacks; or other changes in mood or behavior; suicidal thoughts or actions
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection (IV):

- Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT

include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Apo diazepam | Diazepam | Stesolid | Valium;
- (AR) Argentina: Corticosan | Cuadel | Dezepan | Diactal | Diazepam fabra | Diazepam ingram | Diazepam klonal | Diazepam Lan | Diazepam lepret | Diazepam luar | Diazepam medifarma | Diazepam medipharma | Diazepam northia | Diazepam Puntanos | Diazepam vannier | Diazepam biocrom | Diazepam bouzen | Diazepam fabop | Diazepam fabra | Diazepam vannier | Dipezona | Dipezona plus | Fabotranil | Glutasedan | Lembrol | Plidan | Psiconor | Psicotop | Saromet | Somasedan | Valium | Valtoco;
- (AT) Austria: Gewacalm | Psychopax | Umbrum | Valium;
- (AU) Australia: Antenex | Apo diazepam | Apx diazepam | Diazepam arx | Diazepam dp | Diazepam Ga | Diazepam str | Ducene | Noumed diazepam | Ranzepam | Valium | Valpam;
- (BD) Bangladesh: Azepam | D pam | Diason | Diazimet | Dipam | Easium | Evalin | Fizepam | Relaxen | Rozam | Sedapen | Sedex | Sedil | Sedium | Sedulin | Seduxen | Selium | Strenil | Sudex | Therapam | Tranquil;
- (BE) Belgium: Diazepam EG | Diazepam eurogenerics | Diazepam ratiopharm | Diazepam teva | Diazepam teva generics belgium | Diazetop | Valium;
- (BG) Bulgaria: Dizopin | Relanium | Valium;
- (BR) Brazil: Calmociteno | Compaz | Diazefast | Diazepam nq | Diazepam | Diazepam nq | Dienpax | Dienzepax | Furp diazepam | Kiatrium | Lfm diazepam | Menostress | Noan | Oxatrat | Relapax | Santiazepam | Somaplus | Uni diazepam | Valium | Valix;
- (CH) Switzerland: Diazepam desitin | Diazepam mepha | Paceum | Stesolid | Valium;
- (CL) Chile: Bamyl | Compaz | Elongal | Pacinax | Valium;
- (CO) Colombia: Clafer | Diazepam | Distensar | Valium;
- (CR) Costa Rica: Diazepam mk | Tecepan;
- (CZ) Czech Republic: Apo diazepam | Diazepam biotika | Diazepam desitin | Diazepam stada;
- (DE) Germany: Diazep | Diazep CT | Diazepam stada | Faustan | Lamra | Neurolytril | Tranquase | Valium;
- (DO) Dominican Republic: Calmaven | Diazepam | Diazex | Distenol | Valium | Vatron | Vaxepan;
- (EC) Ecuador: Ansan | Consilium | Diazepam Apo | Diazepam Ecar | Dipaz | Valium;
- (EE) Estonia: Apaurin | Diapaks | Diazepam accord | Diazepam desitin | Diazepam ratiopharm | Diazepex | Faustan | Relium | Seduxen | Singlepen d | Valium;
- (EG) Egypt: Calium | Epival | Farcozepam | Neuril | Stesolid | Valinil | Valpam;
- (ES) Spain: Diaceplex | Diazepam apotex | Diazepam bluefish | Diazepam cinfa | Diazepam durban | Diazepam grindeks | Diazepam lesvi | Diazepam normogen | Diazepam pensa | Diazepam qualigen | Diazepam stada | Diazepam vir | Diazepam normon | Diazepam prodes | Drenian | Valium;
- (FI) Finland: Apozepam | Diapam | Diazal | Medipam | Sedipam | Stesolid | Tensopam | Valium;
- (FR) France: Diazepam accord | Diazepam arrow | Diazepam Merckle | Diazepam ratiopharm | Novazam | Valium;
- (GB) United Kingdom: Alupram | Atensine | Dialar | Diazemuls | Diazepam almus | Diazepam arrow | Diazepam astec | Diazepam berk | Diazepam cox | Diazepam kent | Diazepam sandoz | Evacalm | Rimapam | Sedapam | Stesolid | Tensium | Valium;
- (GR) Greece: Apollonset | Atarviton | Stedon;
- (HK) Hong Kong: Kratium | Sedapam | Syn diazepam | Synium | Valencia | Valium;

- (HR) Croatia: Apaurin | Diazepam Alkaloid | Diazepam jadran | Diazepan Jadran | Normabel;
- (HU) Hungary: Diazepam desitin | Seduxen | Valium;
- (ID) Indonesia: Decazepam | Diazepin | Lovium | Mentalium | Paralium | Prozepam | Sakazepam | Stesolid | Trankinon | Valdimex | Validex | Valinter | Valisanbe | Valium;
- (IE) Ireland: Anxicalm | Diazemuls | Valium;
- (IL) Israel: Assival | Disopam;
- (IN) India: Alvopam | Anxol | Calmpose | Daize | Dekopam | Diacalm | Diaze | Dipax | Direc | Dizep | Elpam | Equimood | Lakpam | Medipam | Neo mesmar | Paci quil | Paxum | Peacin | Placidox | Quietal | Repam | Valium | Zecalm | Zepase | Zepose | Zycalm;
- (IQ) Iraq: Diapam | Diazedain | Diazepam mdi | Diazepam sobhan | Dij valom | Valiapam | Valiupam | Valum j;
- (IT) Italy: Ansiolin | Diazepam stada | Diazepam vm | Micropam | Noan | Valium | Vatran;
- (JO) Jordan: Remedium | Stedon | Stesolid | Valium;
- (JP) Japan: Amel mita | Cellkam | Cercine | Condition | Curasina | Diapax | Diazepam amel | Diazepam fujimoto | Diazepam mita | Diazepam sawai | Diazepam taiho | Diazepam towa | Diazepam tsuruhara | Euphorin | Gihitan | Horizon | Pealkit | Reliezen | Reliver | Renborin | Sedaril | Serenamin | Serenzin | Sonacon;
- (KE) Kenya: Biozepam | Calm | Cozepam | Da zepam | Valium;
- (KR) Korea, Republic of: Diazepam daewon | Melode | Myungin diazepam | Samjin Diazepam | Unizepam | Valium | Valuzepam;
- (KW) Kuwait: Valium;
- (LB) Lebanon: Diazepan | Stesolid | Valium;
- (LT) Lithuania: Apaurin | Apo diazepam | Diapax | Diazepam d | Diazepex | Faustan | Relanium | Relium | Seduxen | Sibazon | Valium;
- (LU) Luxembourg: Valium;
- (LV) Latvia: Apaurin | Apo diazepam | Diapax | Diazep | Diazepam desitin | Diazepex | Diazer | Faustan | Relanium | Relium | Seduxen | Sibazon | Valium;
- (MA) Morocco: Apo diazepam | Valium;
- (MX) Mexico: Alboral | Freudal | Ifafonal | Nercon | Ortopsique | Rayne | Relazepam | Valium;
- (MY) Malaysia: Apo diazepam | Calmpose | Diapine | Diapo | Dzp | Idaman pharma diazepam | Stesolid | Valium;
- (NG) Nigeria: Ex5 diazepam | Unival | Valinex | Vincopam;
- (NL) Netherlands: Diazepam accord | Diazepam alpharma | Diazepam aurobindo | Diazepam flx | Diazepam Katwijk | Diazepam mylan | Diazepam ratiopharm | Diazepam sandoz | Diazepamum | Stesolid | Valium;
- (NO) Norway: Diazepam accord | Stesolid | Valium | Vival;
- (NZ) New Zealand: D pam | Noumed diazepam | Propam | Stesolid;
- (PA) Panama: Ansiopan | Tecepan;
- (PE) Peru: Ansiomed | Apo-diazepan | Pacitran | Reposepam | Valium | Vazen;
- (PH) Philippines: Anxionil | Pamizep | Trazepam | Tremodil | Ul diazepam | Valium | Valzep;
- (PK) Pakistan: Anglopam | Anxosal | Apaurin | Arzepam | Cerelium | D mad | Delica | Depam | Diazip | Dizep | Kozypam | Neopam | Relax | Relaxipam | Rorpam | Tdpam | Valium;
- (PL) Poland: Diazepam genoptim | Diazepam grindeks | Faustan | Neorelium | Relanium | Stesolid | Valium;
- (PR) Puerto Rico: Diastat | Diastat acudial | Libervant | Valium | Valtoco;
- (PT) Portugal: Bialzepam | Diazepam bluefish | Diazepam grindeks | Diazepam Labesfal | Diazepam Pharmakern | Diazepam ratiopharm | Diazepam wynn | Metamidol | Stesolid | Unisedil | Valium;
- (PY) Paraguay: Aneurol | Ansedan | Diazepan prodes | Lembrol | Psicozepam | Rolasedan | Valium;
- (QA) Qatar: Antenex | Diaksi | Diapam | Diazem | Epival (EIPICO) | Stesolid | Valium | Valtoco | Verzepam;
- (RO) Romania: Diazepam desitin | Diazepam laropharm | Diazepam richter | Diazepam terapia | Valium;
- (RU) Russian Federation: Apaurin | Apo diazepam | Diapam | Diazepam desitin | Diazepex | Relanium | Relium | Seduxen | Sibazon | Sibazon ferein | Valium;
- (SA) Saudi Arabia: Stesolid | Valium;

- (SE) Sweden: Apozepam | Diazemuls novum | Diazepam accord | Diazepam bbs | Diazepam Nordic | Diazepam orifarm | Diazepam pilum | Stesolid | Stesolid novum | Valium;
- (SG) Singapore: Apo diazepam | Diapine | Diapo | Stesolid;
- (SI) Slovenia: Apaurin;
- (SK) Slovakia: Diazepam desitin | Diazepam grindeks | Diazepam slovakofarma;
- (SR) Suriname: Apo diazepam;
- (TH) Thailand: A Zerax | Ammipam | Diadon | Diamed | Diano | Diapam | Diapine | Diaz | Diaza | Dinazep | Dipam | Divopan | Dizan | Dizep | Dizepam | Dopam | Dzp | Manodiazepam | Manolium | Maya | Popam | Relanium | Ropam | Seduxen | Sipam | Stesolid | Tranolan | V day zepam | Valax | Valenium | Valium | Vapam | Vason | Zam | Zepam | Zopam;
- (TN) Tunisia: Remedium | Valium;
- (TR) Turkey: Anksiyolin | Diapam | Diazepam | Nervium | Zepam;
- (TW) Taiwan: Anlin | Azepin | Balidium | Baogin | Bayu | Colsin | Comfort | Cozepam | Dean | Derginpin | Dialium | Dianlin | Dianzem | Diapin | Diapine | Diazejin | Diazelium | Dizepam | Dupin | Editin | Ginbo | Hua Pam | Jinlun | Jinpinfan | Neo Zine | Recin | Relax | Sedonin | Sonacon | Soudipan | Stesolid | Sulin | Talium | Tening | Thinin | Toufong | Tranzepam | Valisin | Valium | Vancolin | Vanconin | Vandrin | Walin | Wingent | Winii;
- (UA) Ukraine: Apaurin | Diapam | Diazepam Apo | Relanium | Relium | Seduxen | Sibazon | Sibazone;
- (UG) Uganda: Cozepam | Da zepam | Solina;
- (UY) Uruguay: Compaz | Diazepam icu vita | Diazepam Sanderson | Diazepam | Glorius | Quiatril | Sincronex | Val | Valium | Valzepam;
- (VE) Venezuela, Bolivarian Republic of: Diazepam | Talema;
- (ZA) South Africa: Benzopin | Betapam | Dynapam | Pax | Rolab-diazepam | Valium;
- (ZM) Zambia: Dizam;
- (ZW) Zimbabwe: Dizam

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REFERENCES

1. 2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc.* 2023;71(7):2052-2081. doi:10.1111/jgs.18372 [PubMed 37139824]
2. Abrams GM, Wakasa M. Chronic complications of spinal cord injury and disease. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 25, 2025.
3. Ahlfors CE. Benzyl alcohol, kernicterus, and unbound bilirubin. *J Pediatr.* 2001;139(2):317-319. doi: 10.1067/mpd.2001.116281. [PubMed 11487763]
4. Akerele E, Olupona T. Drugs of abuse. *Psychiatr Clin North Am.* 2017;40(3):501-517. doi: 10.1016/j.psc.2017.05.006 [PubMed 28800805]
5. American Academy of Pediatrics; American Academy of Pediatric Dentistry, Coté CJ, Wilson S; Work Group on Sedation. Guidelines for monitoring and management of pediatric patients during and after sedation for diagnostic and therapeutic procedures: an update. *Pediatrics.* 2006;118(6):2587-2602. doi: 10.1542/peds.2006-2780. [PubMed 17142550]
6. American College of Obstetricians and Gynecologists (ACOG). Gynecologic management of adolescents and young women with seizure disorders: ACOG Committee Opinion, Number 806. *Obstet Gynecol.* 2020;135(5):e213-e220. doi:10.1097/AOG.0000000000003827 [PubMed 32332416]
7. American College of Obstetricians and Gynecologists (ACOG). Treatment and management of mental health conditions during pregnancy and postpartum: ACOG Clinical Practice Guideline No. 5. *Obstet Gynecol.* 2023;141(6):1262-1288. doi:10.1097/AOG.0000000000005202 [PubMed 37486661]

8. American Society of Addiction Medicine (ASAM). The ASAM clinical practice guideline on alcohol withdrawal management. *J Addict Med*. 2020;14(3S)(suppl 1):1-72. doi:10.1097/ADM.0000000000000668 [PubMed 32511109]
9. Anderson PO, Sauberan JB. Modeling drug passage into human milk. *Clin Pharmacol Ther*. 2016;100(1):42-52. [PubMed 27060684]
10. Andrade C, Varadharajan N, Bascarane S, Menon V. Gestational exposure to benzodiazepines or z-hypnotics and neurodevelopmental disorders in offspring: systematic review and meta-analysis. *Acta Psychiatr Scand*. 2024;150(2):65-77. doi:10.1111/acps.13696 [PubMed 38751163]
11. Arnold TC, Ryan ML. Acute amphetamine and synthetic cathinone ("bath salt") intoxication. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 29, 2019.
12. Arroliga AC, Shehab N, McCarthy K, Gonzales JP. Relationship of continuous infusion lorazepam to serum propylene glycol concentration in critically ill adults. *Crit Care Med*. 2004;32(8):1709-1714. [PubMed 15286548]
13. Bandelow B, Allgulander C, Baldwin DS, et al. World Federation of Societies of Biological Psychiatry (WFSBP) guidelines for treatment of anxiety, obsessive-compulsive and posttraumatic stress disorders - version 3. Part I: anxiety disorders. *World J Biol Psychiatry*. 2023;24(2):79-117. doi:10.1080/15622975.2022.2086295 [PubMed 35900161]
14. Bankhead R, Boullata J, Brantley S, et al; A.S.P.E.N. Board of Directors. Enteral nutrition practice recommendations. *JPEN J Parenter Enteral Nutr*. 2009;33(2):122-167. doi:10.1177/0148607108330314 [PubMed 19171692]
15. Barnes BJ, Gerst C, Smith JR, Terrell AR, Mullins ME. Osmol gap as a surrogate marker for serum propylene glycol concentrations in patients receiving lorazepam for sedation. *Pharmacotherapy*. 2006;26(1):23-33. [PubMed 16422667]
16. Barry JD. Antimalarials. In: Nelson LS, Howland M, Lewin NA, Smith SW, Goldfrank LR, Hoffman RS, eds. *Goldfrank's Toxicologic Emergencies*. 11th ed. New York, NY: McGraw-Hill; 2019:836-849.
17. Basura GJ, Adams ME, Monfared A, et al. Clinical practice guideline: Ménière's disease. *Otolaryngol Head Neck Surg*. 2020;162(suppl 2):S1-S55. doi:10.1177/0194599820909438 [PubMed 32267799]
18. Becker DA, Wheless JW, Sirven J, Tatum WO, Rabinowicz AL, Carrazana E. Treatment of seizure clusters in epilepsy: a narrative review on rescue therapies. *Neurol Ther*. 2023;12(5):1439-1455. doi:10.1007/s40120-023-00515-3 [PubMed 37341903]
19. Bellantuono C, Tofani S, Di Sciascio G, Santone G. Benzodiazepine exposure in pregnancy and risk of major malformations: a critical overview. *Gen Hosp Psychiatry*. 2013;35(1):3-8. doi:10.1016/j.genhosppsy.2012.09.003 [PubMed 23044244]
20. Bhattacharyya N, Gubbels SP, Schwartz SR, et al. Clinical practice guideline: Benign paroxysmal positional vertigo (update). *Otolaryngol Head Neck Surg*. 2017;156(3)(suppl):S1-S47. doi:10.1177/0194599816689667. [PubMed 28248609]
21. Bird S. Organophosphate and carbamate poisoning. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed September 26, 2025.
22. Boullata JL, Carrera AL, Harvey L, et al. ASPEN safe practices for enteral nutrition therapy. *JPEN J Parenter Enteral Nutr*. 2017;41(1):15-103. doi:10.1177/0148607116673053 [PubMed 27815525]
23. Boullata JL. *Guidebook on Enteral Medication Administration*. American Society for Parenteral and Enteral Nutrition; 2019.

24. Boyer EW. Serotonin syndrome (serotonin toxicity). Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 7, 2020.
25. Boyer EW. Methamphetamine: Acute intoxication. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 16, 2019b.
26. Brandt R. Passage of diazepam and desmethyldiazepam into breast milk. *Arzneimittelforschung*. 1976;26(3):454-457. [PubMed 989345]
27. Brophy GM, Bell R, Claassen J, et al; Neurocritical Care Society Status Epilepticus Guideline Writing Committee. Guidelines for the evaluation and management of status epilepticus. *Neurocrit Care*. 2012;17(1):3-23. doi:10.1007/s12028-012-9695-z [PubMed 22528274]
28. Centers for Disease Control and Prevention (CDC). Neonatal deaths associated with use of benzyl alcohol—United States. *MMWR Morb Mortal Wkly Rep*. 1982;31(22):290-291. <http://www.cdc.gov/mmwr/preview/mmwrhtml/00001109.htm> [PubMed 6810084]
29. Chiang LM, Wang HS, Shen HH, et al. Rectal diazepam solution is as good as rectal administration of intravenous diazepam in the first-aid cessation of seizures in children with intractable epilepsy. *Pediatr Neonatol*. 2011;52(1):30-33. doi:10.1016/j.pedneo.2010.12.009 [PubMed 21385654]
30. Choy Y. Acute procedural anxiety and specific phobia of clinical procedures in adults: Treatment overview. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 2, 2019.
31. Chuang HM, Meng LC, Lin CW, et al. Concomitant use of antidepressants and benzodiazepines during pregnancy and associated risk of congenital malformations: a population-based cohort study in Taiwan. *Lancet Psychiatry*. 2024;11(8):601-610. doi:10.1016/S2215-0366(24)00176-7 [PubMed 38968942]
32. Clark E, Giambra BK, Hingl J, Doellman D, Tofani B, Johnson N. Reducing risk of harm from extravasation: a 3-tiered evidence-based list of pediatric peripheral intravenous infusates. *J Infus Nurs*. 2013;36(1):37-45. doi:10.1097/NAN.0b013e3182798844 [PubMed 23271150]
33. Cloyd JC, Lalonde RL, Beniak TE, et al. A single-blind, crossover comparison of the pharmacokinetics and cognitive effects of a new diazepam rectal gel with intravenous diazepam. *Epilepsia*. 1998;39(5):520-526. [PubMed 9596205]
34. Cole AP, Hailey DM. Diazepam and active metabolite in breast milk and their transfer to the neonate. *Arch Dis Child*. 1975;50(9):741-742. [PubMed 1190825]
35. Davanzo R, Dal Bo S, Bua J, Copertino M, Zanelli E, Matarazzo L. Antiepileptic drugs and breastfeeding. *Ital J Pediatr*. 2013;39:50. [PubMed 23985170]
36. Davidoff K. Pediatric palliative care. In: Kliegman RM, St. Geme J, eds. *Nelson Textbook of Pediatrics*. 22nd ed. Elsevier; 2024:chap. 8.
37. Delgado J. Intoxication from LSD and other common hallucinogens. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 7, 2020.
38. Diastat and Diastat AcuDial (diazepam) [prescribing information]. Bridgewater, NJ: Bausch Health US LLC; July 2023.
39. Diazepam carpject [prescribing information]. Lake Forest, IL: Hospira; August 2022.
40. Diazepam injection [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA, Inc; May 2023.
41. Diazepam injection multi-dose vial [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; May 2023.
42. Diazepam injection prefilled syringe [prescribing information]. Upper Saddle River, NJ: DASH Pharmaceuticals, LLC; March 2021.
43. Diazepam intramuscular autoinjector [prescribing information]. Columbia, MD: Meridian Medical Technologies; July 2005.

44. Diazepam oral solution and diazepam oral solution concentrate [prescribing information]. Congers, NY: Chartwell RX; July 2024.
45. Dickson EW, Bird SB, Gaspari RJ, Boyer EW, Ferris CF. Diazepam inhibits organophosphate-induced central respiratory depression. *Acad Emerg Med*. 2003;10(12):1303-1306. doi:10.1111/j.1553-2712.2003.tb00001.x [PubMed 14644779]
46. Dolder CR, Nelson MH. Hypnotic-induced complex behaviours: incidence, mechanisms and management. *CNS Drugs*. 2008;22(12):1021-1036. doi:10.2165/0023210-200822120-00005. [PubMed 18998740]
47. Drappatz J, Avila EK. Seizures in patients with primary and metastatic brain tumors. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 19, 2019.
48. Drislane FW. Convulsive status epilepticus in adults: Management. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed February 13, 2025.
49. Dusci LJ, Good SM, Hall RW, Ilett KF. Excretion of diazepam and its metabolites in human milk during withdrawal from combination high dose diazepam and oxazepam. *Br J Clin Pharmacol*. 1990;29(1):123-126. [PubMed 2105100]
50. Eleftheriou G, Zandonella Callegher R, et al. Consensus panel recommendations for the pharmacological management of breastfeeding women with postpartum depression. *Int J Environ Res Public Health*. 2024;21(5):551. doi:10.3390/ijerph21050551 [PubMed 38791766]
51. Engle HA, "The Effect of Diazepam (Valium) in Children With Cerebral Palsy: A Double-Blind Study," *Dev Med Child Neurol*, 1966, 8(6):661-67. [PubMed 5339679]
52. Erkkola R, Kanto J. Diazepam and breast-feeding. *Lancet*. 1972;1(7762):1235-1236. [PubMed 4113217]
53. Everitt IJ, Barnett P. Comparison of two benzodiazepines used for sedation of children undergoing suturing of a laceration in an emergency department. *Pediatr Emerg Care*. 2002;18(2):72-74. [PubMed 11973494]
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56. Fell D, Gough MB, Northan AA, Henderson CU. Diazepam premedication in children. Plasma levels and clinical effects. *Anaesthesia*. 1985;40(1):12-17. [PubMed 3970333]
57. Francois D, Roberts J, Hess S, Probst L, Eksioğlu Y. Medical management with diazepam for electrical status epilepticus during slow wave sleep in children. *Pediatr Neurol*. 2014;50(3):238-242. doi:10.1016/j.pediatrneurol.2013.11.002 [PubMed 24393416]
58. Freeman MP, Góez-Mogollón L, McInerney KA, et al. Obstetrical and neonatal outcomes after benzodiazepine exposure during pregnancy: results from a prospective registry of women with psychiatric disorders. *Gen Hosp Psychiatry*. 2018;53:73-79. doi:10.1016/j.genhosppsych.2018.05.010 [PubMed 29958100]
59. Furman JM, Barton JJS. Treatment of vertigo. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 28, 2022.
60. Gilder ME, Tun NW, Carter A, et al. Outcomes for 298 breastfed neonates whose mothers received ketamine and diazepam for postpartum tubal ligation in a resource-limited setting. *BMC Pregnancy Childbirth*. 2021;21(1):121. doi:10.1186/s12884-021-03610-1 [PubMed 33563234]

61. Ginsberg GG, Lewis JH, Gallagher JE, et al. Diazepam versus midazolam for colonoscopy: a prospective evaluation of predicted versus actual dosing requirements. *Gastrointest Endosc.* 1992;38(6):651-656. doi: 10.1016/s0016-5107(92)70559-8 [PubMed 1473667]
62. Glauser T, Shinnar S, Gloss D, et al. Evidence-based guideline: treatment of convulsive status epilepticus in children and adults: report of the Guideline Committee of the American Epilepsy Society. *Epilepsy Curr.* 2016;16(1):48-61. doi:10.5698/1535-7597-16.1.48 [PubMed 26900382]
63. Greenblatt DJ, Ehrenberg BL, Gunderman J, et al. Pharmacokinetic and electroencephalographic study of intravenous diazepam, midazolam, and placebo. *Clin Pharmacol Ther.* 1989a;45(4):356-365. [PubMed 2702793]
64. Greenblatt DJ, Harmatz JS, Friedman H, Locniskar A, Shader RI. A large-sample study of diazepam pharmacokinetics. *Ther Drug Monit.* 1989b;11(6):652-657. [PubMed 2512694]
65. Greenblatt DJ, Koch-Weser J. Clinical toxicity of chlordiazepoxide and diazepam in relation to serum albumin concentration: a report from the Boston Collaborative Drug Surveillance Program. *Eur J Clin Pharmacol.* 1974;7(4):259-262. doi:10.1007/BF00560341 [PubMed 4851053]
66. Griffin CE 3rd, Kaye AM, Bueno FR, Kaye AD. Benzodiazepine pharmacology and central nervous system-mediated effects. *Ochsner J.* 2013;13(2):214-223. [PubMed 23789008]
67. Grigoriadis S, Graves L, Peer M, et al. Benzodiazepine use during pregnancy alone or in combination with an antidepressant and congenital malformations: systematic review and meta-analysis. *J Clin Psychiatry.* 2019;80(4):18r12412. doi:10.4088/JCP.18r12412 [PubMed 31294935]
68. Hain TC, Uddin M. Pharmacological treatment of vertigo. *CNS Drugs.* 2003;17(2):85-100. doi: 10.2165/00023210-200317020-00002 [PubMed 12521357]
69. Hall WC, Talbert RL, Ereshefsky L. Cocaine abuse and its treatment. *Pharmacotherapy.* 1990;10(1):47-65. [PubMed 2179901]
70. Hoffman RS. Management of moderate and severe alcohol withdrawal syndromes. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 2, 2023.
71. Holt KS. The use of diazepam in childhood cerebral palsy. Report of a small study including electromyographic observations. *Ann Phys Med.* 1964;Suppl:16-24. [PubMed 5317889]
72. Holt SR, Tetrault JM. Alcohol withdrawal: Ambulatory management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 2, 2023.
73. Hurst S, McMillan M. Innovative solutions in critical care units: extravasation guidelines. *Dimens Crit Care Nurs.* 2004;23(3):125-128. doi:10.1097/00003465-200405000-00009 [PubMed 15192356]
74. "Inactive" ingredients in pharmaceutical products: update (subject review). American Academy of Pediatrics Committee on Drugs. *Pediatrics.* 1997;99(2):268-278. doi: 10.1542/peds.99.2.268. [PubMed 9024461]
75. Iqbal MM, Sobhan T, Ryals T, et al, "Effects of Commonly Used Benzodiazepines on the Fetus, the Neonate, and the Nursing Infant," *Psychiatr Serv*, 2002, 53(1):39-49. [PubMed 11773648]
76. Ito S. Drug therapy for breast-feeding women. *N Engl J Med.* 2000;343(2):118-126. [PubMed 10891521]
77. Jennum P, Baandrup L, Ibsen R, Kjellberg J. Increased all-cause mortality with use of psychotropic medication in dementia patients and controls: a population-based register study. *Eur Neuropsychopharmacol.* 2015;25(11):1906-1913. doi: 10.1016/j.euroneuro.2015.08.014. [PubMed 26342397]
78. Kanto JH. Use of benzodiazepines during pregnancy, labour and lactation, with particular reference to pharmacokinetic considerations. *Drugs.* 1982;23(5):354-380. doi:10.2165/00003495-198223050-00002 [PubMed 6124415]

79. Katzman MA, Bleau P, Blier P, Chokka P, Kjernisted K, Van Ameringen M; Canadian Anxiety Guidelines Initiative Group on behalf of the Anxiety Disorders Association of Canada/Association Canadienne des Troubles Anxieux and McGill University. Canadian clinical practice guidelines for the management of anxiety, posttraumatic stress and obsessive-compulsive disorders. *BMC Psychiatry*. 2014;14(suppl 1):S1. doi: 10.1186/1471-244X-14-S1-S1 [PubMed 25081580]
80. Kishimoto S, Nakamura K, Arai T, Yukimasa O, Fukami N. Postoperative neuroleptic malignant syndrome-like symptoms improved with intravenous diazepam: a case report. *J Anesth*. 2013;27(5):768-770. doi: 10.1007/s00540-013-1602-z [PubMed 23553148]
81. Kita M, Goodkin DE. Drugs used to treat spasticity. *Drugs*. 2000;59(3):487-495. doi: 10.2165/00003495-200059030-00006 [PubMed 10776831]
82. Krauss B, Green SM. Procedural sedation and analgesia in children. *Lancet*. 2006;367(9512):766-780. [PubMed 16517277]
83. Kuang H, Li Y, Lu Y, Zhang L, Wei L, Wu Y. Reproductive and fetal outcomes in women with epilepsy: a systematic review and meta-analysis. *J Matern Fetal Neonatal Med*. 2024;37(1):2351196. doi:10.1080/14767058.2024.2351196 [PubMed 38735863]
84. Lader M. Benzodiazepines revisited--will we ever learn? *Addiction*. 2011;106(12):2086-2109. doi:10.1111/j.1360-0443.2011.03563.x [PubMed 21714826]
85. Lamson MJ, Sitki-Green D, Wannarka GL, Mesa M, Andrews P, Pellock J. Pharmacokinetics of diazepam administered intramuscularly by autoinjector versus rectal gel in healthy subjects: a phase I, randomized, open-label, single-dose, crossover, single-centre study. *Clin Drug Investig*. 2011;31(8):585-597. [PubMed 21721594]
86. Lavonas EJ, Akpunonu PD, Arens AM, et al. 2023 American Heart Association focused update on the management of patients with cardiac arrest or life-threatening toxicity due to poisoning: an update to the American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care. *Circulation*. 2023;148(16):e149-e184. doi:10.1161/CIR.0000000000001161 [PubMed 37721023]
87. Leppik IE, Patel SI. Intramuscular and rectal therapies of acute seizures. *Epilepsy Behav*. 2015;49:307-312. doi: 10.1016/j.yebeh.2015.05.001 [PubMed 26071998]
88. Libervant (diazepam) [prescribing information]. Warren, NJ: Aquestive Therapeutics; April 2024.
89. Ling Ngan Wong A, Tsz Fung Cheung I, Graham CA. Hydroxychloroquine overdose: case report and recommendations for management. *Eur J Emerg Med*. 2008;15(1):16-18. doi:10.1097/MEJ.0b013e3280adcb56 [PubMed 18180661]
90. Lott E, Fehlings D, Gelineau-Morel R, et al. Physician approaches to the pharmacologic treatment of dystonia in cerebral palsy. Preprint. *medRxiv*. 2024;2024.02.01.24302121. doi:10.1101/2024.02.01.24302121 [PubMed 38352331]
91. Mancuso CE, Tanzi MG, Gabay M. Paradoxical reactions to benzodiazepines: literature review and treatment options. *Pharmacotherapy*. 2004;24(9):1177-1185. doi: 10.1592/phco.24.13.1177.38089. [PubMed 15460178]
92. Marquardt K, Albertson TE. Treatment of hydroxychloroquine overdose. *Am J Emerg Med*. 2001;19(5):420-424. doi:10.1053/ajem.2001.25774 [PubMed 11555803]
93. Martinez JL, Sutters KA, Waite S, et al. A comparison of oral diazepam versus midazolam, administered with intravenous meperidine, as premedication to sedation for pediatric endoscopy. *J Pediatr Gastroenterol Nutr*. 2002;35(1):51-58. [PubMed 12142810]
94. Mathew A, Mathew MC, Thomas M, Antonisamy B. The efficacy of diazepam in enhancing motor function in children with spastic cerebral palsy. *J Trop Pediatr*. 2005;51(2):109-113. doi:10.1093/tropej/fmh095 [PubMed 15840761]
95. Mazzone PP, Hogg KM, Weir CJ, Stephen J, Bhattacharya S, Chin RFM. Comparison of perinatal outcomes for women with and without epilepsy: a systematic review and meta-analysis. *JAMA*

Neurol. 2023;80(5):484-494. doi:10.1001/jamaneurol.2023.0148 [PubMed 36912826]

96. McAllister-Williams RH, Baldwin DS, Cantwell R, et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. *J Psychopharmacol.* 2017;31(5):519-552. doi:10.1177/0269881117699361 [PubMed 28440103]
97. McBeth PB, Missirlis PI, Brar H, Dhingra V. Novel therapies for myocardial irritability following extreme hydroxychloroquine toxicity. *Case Rep Emerg Med.* 2015;2015:692948. doi:10.1155/2015/692948 [PubMed 26351590]
98. McDonough JH Jr, Zoeffel LD, McMonagle J, Copeland TL, Smith CD, Shih TM. Anticonvulsant treatment of nerve agent seizures: anticholinergics versus diazepam in soman-intoxicated guinea pigs. *Epilepsy Res.* 2000;38(1):1-14. doi:10.1016/s0920-1211(99)00060-1 [PubMed 10604601]
99. McElhatton PR. The effects of benzodiazepine use during pregnancy and lactation. *Reprod Toxicol.* 1994;8(6):461-475. [PubMed 7881198]
100. Mikati MA, Tchapyjnikov D, Rathke KM. Febrile seizures. In: Kliegman RM, St. Geme J, eds. *Nelson Textbook of Pediatrics.* 22nd ed. Elsevier; 2024:chap. 633.1.
101. Milligan NM, Dhillon S, Griffiths A, Oxley J, Richens A. A clinical trial of single dose rectal and oral administration of diazepam for the prevention of serial seizures in adult epileptic patients. *J Neurol Neurosurg Psychiatry.* 1984;47(3):235-240. doi:10.1136/jnnp.47.3.235 [PubMed 6368753]
102. Milsap RL, Jusko WJ. Pharmacokinetics in the infant. *Environ Health Perspect.* 1994;102 Suppl 11:107-110. [PubMed 7737034]
103. Moore TA. Ambulatory detoxification in alcohol use disorder and opioid use disorder. *Ment Health Clin.* 2020;10(6):307-316. doi:10.9740/mhc.2020.11.307 [PubMed 33224688]
104. Morselli PL, Franco-Morselli R, Bossi L. Clinical pharmacokinetics in newborns and infants. Age-related differences and therapeutic implications. *Clin Pharmacokinet.* 1980;5(6):485-527.
105. Morselli PL, Principi N, Tognoni G, et al. Diazepam elimination in premature and full term infants, and children. *J Perinat Med.* 1973;1(2):133-141.
106. Moskowitz HS, Dinces EA. Meniere disease. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 2, 2022.
107. Murphy MR, Blick DW, Dunn MA, Fanton JW, Hartgraves SL. Diazepam as a treatment for nerve agent poisoning in primates. *Aviat Space Environ Med.* 1993;64(2):110-115. [PubMed 8431183]
108. National Institute for Health and Care Excellence (NICE). Epilepsies in children, young people and adults. <https://www.nice.org.uk/guidance/ng217>. Published April 27, 2022. Accessed June 19, 2024.
109. Nelson J, Chouinard G. Guidelines for the clinical use of benzodiazepines: pharmacokinetics, dependency, rebound and withdrawal. Canadian Society for Clinical Pharmacology. *Can J Clin Pharmacol.* 1999;6(2):69-83. [PubMed 10519733]
110. Ochs HR, Greenblatt DJ, Kaschell HJ, Klehr U, Divoll M, Abernethy DR. Diazepam kinetics in patients with renal insufficiency or hyperthyroidism. *Br J Clin Pharmacol.* 1981;12(6):829-832. doi:10.1111/j.1365-2125.1981.tb01315.x [PubMed 7340884]
111. Ochs HR, Rauh HW, Greenblatt DJ, Kaschell HJ. Clorazepate dipotassium and diazepam in renal insufficiency: serum concentrations and protein binding of diazepam and desmethyldiazepam. *Nephron.* 1984;37(2):100-104. doi:10.1159/000183222 [PubMed 6145109]
112. Olek MJ, Narayan RN, Frohman EM, Frohman TC. Symptom management of multiple sclerosis in adults. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed July 6, 2022.
113. Oleszek JL, Davidson LT. Spasticity. In: Kliegman RM, St. Geme J, eds. *Nelson Textbook of Pediatrics.* 22nd ed. Elsevier; 2024:chap. 752.

114. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med.* 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y [PubMed 37256475]
115. Pack AM, Oskoui M, Williams Roberson S, et al. Teratogenesis, perinatal, and neurodevelopmental outcomes after in utero exposure to antiseizure medication: practice guideline from the AAN, AES, and SMFM. *Neurology.* 2024;102(11):e209279. doi:10.1212/WNL.0000000000209279 [PubMed 38748979]
116. Palmaro A, Dupouy J, Lapeyre-Mestre M. Benzodiazepines and risk of death: results from two large cohort studies in France and UK. *Eur Neuropsychopharmacol.* 2015;25(10):1566-1577. doi:10.1016/j.euroneuro.2015.07.006 [PubMed 26256008]
117. Park TW. Benzodiazepine use disorder. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 12, 2022.
118. Patrick MJ, Tilstone WJ, Reavey P. Diazepam and breast-feeding. *Lancet.* 1972;1(7749):542-543. [PubMed 4110044]
119. Peck J, Urits I, Crane J, et al. Oral muscle relaxants for the treatment of chronic pain associated with cerebral palsy. *Psychopharmacol Bull.* 2020;50(4 Suppl 1):142-162. [PubMed 33633423]
120. Rabinstein AA. Neuroleptic malignant syndrome. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed May 22, 2025.
121. Radojčić MR, El Marroun H, Miljković B, et al. Prenatal exposure to anxiolytic and hypnotic medication in relation to behavioral problems in childhood: a population-based cohort study. *Neurotoxicol Teratol.* 2017;61:58-65. doi:10.1016/j.ntt.2017.02.005 [PubMed 28259732]
122. Rajiv KR, Radhakrishnan A. Status epilepticus in pregnancy - can we frame a uniform treatment protocol? *Epilepsy Behav.* 2019;101(pt B):106376. doi:10.1016/j.yebeh.2019.06.020 [PubMed 31303443]
123. Refer to manufacturer's labeling.
124. Remy C, Jourdil N, Villemain D, Favel P, Genton P. Intrarectal diazepam in epileptic adults. *Epilepsia.* 1992;33(2):353-358. doi: 10.1111/j.1528-1157.1992.tb02327.x [PubMed 1547766]
125. Riou B, Barriot P, Rimaillho A, Baud FJ. Treatment of severe chloroquine poisoning. *N Engl J Med.* 1988;318(1):1-6. doi:10.1056/NEJM198801073180101 [PubMed 3336379]
126. Robertson CE, Eggers SDZ. Vestibular migraine. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 23, 2021.
127. Rosman NP, Colton T, Labazzo J, et al, "A Controlled Trial of Diazepam Administered During Febrile Illnesses to Prevent Recurrence of Febrile Seizures," *N Engl J Med*, 1993, 329(2):79-84. [PubMed 8510706]
128. Saarelainen L, Tolppanen AM, Koponen M, et al. Risk of death associated with new benzodiazepine use among persons with Alzheimer disease: a matched cohort study. *Int J Geriatr Psychiatry.* 2018;33(4):583-590. doi: 10.1002/gps.4821. [PubMed 29143367]
129. Sánchez Fernández I, Chapman K, Peters JM, et al. Treatment for continuous spikes and waves during sleep (CSWS): survey on treatment choices in North America. *Epilepsia.* 2014;55(7):1099-1108. doi:10.1111/epi.12678 [PubMed 24917485]
130. Sánchez Fernández I, Peters JM, An S, et al. Long-term response to high-dose diazepam treatment in continuous spikes and waves during sleep. *Pediatr Neurol.* 2013;49(3):163-170.e4. doi:10.1016/j.pediatrneurol.2013.04.027 [PubMed 23953953]
131. Seigler RS. The administration of rectal diazepam for acute management of seizures. *J Emerg Med.* 1990;8(2):155-159. doi:10.1016/0736-4679(90)90225-k [PubMed 2193988]

132. Shehab N, Lewis CL, Streetman DD, Donn SM. Exposure to the pharmaceutical excipients benzyl alcohol and propylene glycol among critically ill neonates. *Pediatr Crit Care Med*. 2009;10(2):256-259. [PubMed 19188870]
133. Sheno RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs used to treat pediatric emergencies. *Pediatrics*. 2020;145(1):e20193450. [PubMed 31871244]
134. Shih RD, Walsh B, Eskin B, et al. Diazepam and meclizine are equally effective in the treatment of vertigo: an emergency department randomized double-blind placebo-controlled trial. *J Emerg Med*. 2017;52(1):23-27. doi: 10.1016/j.jemermed.2016.09.016 [PubMed 27789115]
135. Somerville ER, Antony JH. Position statement on the use of rectal diazepam in epilepsy. Epilepsy Society of Australia, the Child Neurology Study Group, the Australian Association of Neurologists, and the National Epilepsy Association of Australia. *Med J Aust*. 1995;163(5):268-269. [PubMed 7565215]
136. Soyka M, Kranzler HR, Hesselbrock V, Kasper S, Mutschler J, Möller HJ; WFSBP Task Force on Treatment Guidelines for Substance Use Disorders. Guidelines for biological treatment of substance use and related disorders, part 1: alcoholism, first revision. *World J Biol Psychiatry*. 2017;18(2):86-119. doi: 10.1080/15622975.2016.1246752 [PubMed 28006997]
137. Steering Committee on Quality Improvement and Management, Subcommittee on Febrile Seizures American Academy of Pediatrics, "Febrile Seizures: Clinical Practice Guideline for the Long-Term Management of the Child With Simple Febrile Seizures," *Pediatrics*, 2008, 121(6):1281-6. [PubMed 18519501]
138. Stefanos SS, Kiser TH, MacLaren R, Mueller SW, Reynolds PM. Management of noncytotoxic extravasation injuries: a focused update on medications, treatment strategies, and peripheral administration of vasopressors and hypertonic saline. *Pharmacotherapy*. 2023;43(4):321-337. doi:10.1002/phar.2794 [PubMed 36938775]
139. Stein MB, Craske M. Generalized anxiety disorder in adults: management. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed September 2, 2025.
140. Stolbach A, Hoffman RS. Opioid withdrawal in adults in the emergency setting. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 1, 2025.
141. Sundbakk LM, Wood M, Gran JM, Nordeng H. Prenatal exposure to benzodiazepine and z-hypnotics and fifth-grade scholastic skills - emulating target trials using data from the Norwegian Mother, Father and Child Cohort Study. *Am J Epidemiol*. 2025;194(1):73-84. doi:10.1093/aje/kwae159 [PubMed 38944758]
142. Thomas SP, Thakkar J, Kovoov P, Thiagalingam A, Ross DL. Sedation for electrophysiological procedures. *Pacing Clin Electrophysiol*. 2014;37(6):781-790. [PubMed 24697803]
143. Thomson FC, Naysmith MR, Lindsay A. Managing drug therapy in patients receiving enteral and parenteral nutrition. *The Pharmaceutical Journal*. <https://pharmaceutical-journal.com/article/ld/managing-drug-therapy-in-patients-receiving-enteral-and-parenteral-nutrition>. Published June 6, 2000. Accessed November 13, 2025. doi:10.1211/PJ.2021.1.65496
144. Tinker SC, Reefhuis J, Bitsko RH, et al; National Birth Defects Prevention Study. Use of benzodiazepine medications during pregnancy and potential risk for birth defects, National Birth Defects Prevention Study, 1997-2011. *Birth Defects Res*. 2019;111(10):613-620. doi:10.1002/bdr2.1497 [PubMed 30891943]
145. Tormoehlen LM, Rusyniak DE. Neuroleptic malignant syndrome and serotonin syndrome. *Handb Clin Neurol*. 2018;157:663-675. doi: 10.1016/B978-0-444-64074-1.00039-2 [PubMed 30459031]
146. Triantafyllidis JK, Merikas E, Nikolakis D, Papalois AE. Sedation in gastrointestinal endoscopy: current issues. *World J Gastroenterol*. 2013;19(4):463-481. [PubMed 23382625]

147. Tsai MC, Huang TL. Severe neuroleptic malignant syndrome: successful treatment with high-dose lorazepam and diazepam: a case report. *Chang Gung Med J*. 2010;33(5):576-580. [PubMed 20979709]
148. Tuovinen K. Organophosphate-induced convulsions and prevention of neuropathological damages. *Toxicology*. 2004;196(1-2):31-39. doi:10.1016/j.tox.2003.10.013 [PubMed 15036754]
149. Tyagi P, Dixit U, Tyagi S, Jain A. Sedative effects of oral midazolam, intravenous midazolam and oral diazepam. *J Clin Pediatr Dent*. 2012;36(4):383-388. [PubMed 23019837]
150. US Department of Veterans Affairs/US Department of Defense. VA/DoD clinical practice guideline for management of posttraumatic stress disorder and acute stress disorder. <https://www.healthquality.va.gov/HEALTHQUALITY/guidelines/mh/ptsd/index.asp>. Published 2023. Accessed November 27, 2023.
151. US Department of Veterans Affairs/US Department of Defense. VA/DoD clinical practice guideline for the management of substance use disorders. <https://www.limbic-cenc.org/wp-content/uploads/VA-DoD-CPG-Management-of-Substance-Use-Disorders-2021-Full-Report.pdf>. Published August 2021. Accessed May 12, 2022.
152. US Food and Drug Administration; Institute for Safe Medication Practices. FDA and ISMP Lists of Look-Alike Drug Names With Recommended Tall Man (Mixed Case) Letters. ISMP; 2023.
153. Valium (diazepam) [prescribing information]. Rahway, NJ: Waylis Therapeutics LLC; August 2024.
154. Valtoco (diazepam intranasal) [prescribing information]. San Diego, CA: Neurelis Inc; June 2025.
155. Vinkers CH, Olivier B. Mechanisms underlying tolerance after long-term benzodiazepine use: a future for subtype-selective GABA(A) receptor modulators? *Adv Pharmacol Sci*. 2012;2012:416864. doi: 10.1155/2012/416864. [PubMed 22536226]
156. Wang RZ, Vashistha V, Kaur S, Houchens NW. Serotonin syndrome: preventing, recognizing, and treating it. *Cleve Clin J Med*. 2016;83(11):810-817. doi: 10.3949/ccjm.83a.15129 [PubMed 27824534]
157. Wang X, Zhang T, Ekheden I, et al. Prenatal exposure to benzodiazepines and Z-drugs in humans and risk of adverse neurodevelopmental outcomes in offspring: a systematic review. *Neurosci Biobehav Rev*. 2022;137:104647. doi:10.1016/j.neubiorev.2022.104647 [PubMed 35367514]
158. Ware MR, Feller DB, Hall KL. Neuroleptic malignant syndrome: diagnosis and management. *Prim Care Companion CNS Disord*. 2018;20(1):17r02185. doi: 10.4088/PCC.17r02185 [PubMed 29325237]
159. Weintraub SJ. Diazepam in the treatment of moderate to severe alcohol withdrawal. *CNS Drugs*. 2017;31(2):87-95. doi: 10.1007/s40263-016-0403-y [PubMed 28101764]
160. Wesson DR, Camber S, Harkey M, Smith DE. Diazepam and desmethyldiazepam in breast milk. *J Psychoactive Drugs*. 1985;17(1):55-56. [PubMed 3920372]
161. White R, Bradnam V; British Pharmaceutical Nutrition Group. *Handbook of Drug Administration via Enteral Feeding Tubes*. 3rd ed. Pharmaceutical Press; 2015.
162. Wichliński LM, Mazur R, Sieradzki E, Jankowski A, Dirska R, Gosk-Kowalska T. Pharmacokinetics of diazepam after intravenous, intramuscular and oral administration to humans. *Mater Med Pol*. 1985;17(1):44-46. [PubMed 4033223]
163. Wightman RS, Nelson LS, Lee JD, Fox LM, Smith SW. Severe opioid withdrawal precipitated by Vivitrol®. *Am J Emerg Med*. 2018;36(6):1128.e1-1128.e2. doi: 10.1016/j.ajem.2018.03.052 [PubMed 29605483]
164. Wilson KC, Reardon C, Farber HW. Propylene glycol toxicity in a patient receiving intravenous diazepam. *N Engl J Med*. 2000;343(11):815. [PubMed 10991709]

165. Wilson KC, Reardon C, Theodore AC, et al. Propylene glycol toxicity: a severe iatrogenic illness in ICU patients receiving IV benzodiazepines: A case series and prospective, observational pilot study. *Chest*. 2005;128(3):1674-1681. [PubMed 16162774]
166. Wodarz N, Krampe-Scheidler A, Christ M, et al. Evidence-based guidelines for the pharmacological management of acute methamphetamine-related disorders and toxicity. *Pharmacopsychiatry*. 2017;50(3):87-95. doi: 10.1055/s-0042-123752 [PubMed 28297728]
167. Wohlt PD, Zheng L, Gunderson S, Balzar SA, Johnson BD, Fish JT. Recommendations for the use of medications with continuous enteral nutrition. *Am J Health Syst Pharm*. 2009;66(16):1458-1467. doi:10.2146/ajhp080632 [PubMed 19667002]
168. World Health Organization, WHO Technical Note. Current recommendations for the treatment of tetanus during humanitarian emergencies. January 2010. <https://www.who.int/publications/i/item/current-recommendations-for-treatment-of-tetanus-during-humanitarian-emergencies>.
169. Wu HN, Liang Y, Li LL, Jiang HY, Xu LL. The safety of benzodiazepines and related drugs during pregnancy: an updated meta-analysis of cohort studies. *Arch Gynecol Obstet*. 2024;310(1):45-54. doi:10.1007/s00404-024-07557-4 [PubMed 38806942]
170. Wusthoff CJ, Shellhaas RA, Licht DJ. Management of common neurologic symptoms in pediatric palliative care: seizures, agitation, and spasticity. *Pediatr Clin North Am*. 2007;54(5):709-xi. [PubMed 17933619]
171. Yahwak JA, Riker RR, Fraser GL, Subak-Sharpe S. Determination of a lorazepam dose threshold for using the osmol gap to monitor for propylene glycol toxicity. *Pharmacotherapy*. 2008;28(8):984-991. [PubMed 18657015]
172. Zakko SF, Seifert HA, Gross JB. A comparison of midazolam and diazepam for conscious sedation during colonoscopy in a prospective double-blind study. *Gastrointest Endosc*. 1999;49(6):684-689. doi: 10.1016/s0016-5107(99)70282-8 [PubMed 10343209]
173. Zar T, Graeber C, Perazella MA. Recognition, treatment, and prevention of propylene glycol toxicity. *Semin Dial*. 2007;20(3):217-219. [PubMed 17555487]
174. Zeltzer LK, Altman A, Cohen D, et al, "Report of the Subcommittee on the Management of Pain Associated With Procedures in Children With Cancer," *Pediatrics*, 1990, 86(5 Pt 2):826-31. [PubMed 2216645]
175. Zhong G, Wang Y, Zhang Y, Zhao Y. Association between benzodiazepine use and dementia: a meta-analysis. *PLoS One*. 2015;10(5):e0127836. doi:10.1371/journal.pone.0127836 [PubMed 26016483]

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